

TODO: Title

This notebook lists all the steps that you need to complete the complete this project. You will need to complete all the TODOs in this notebook as well as in the README and the two python scripts included with the starter code.

TODO: Give a helpful introduction to what this notebook is for. Remember that comments, explanations and good documentation make your project informative and professional.

Note: This notebook has a bunch of code and markdown cells with TODOs that you have to complete. These are meant to be helpful guidelines for you to finish your project while meeting the requirements in the project rubrics. Feel free to change the order of these the TODO's and use more than one TODO code cell to do all your tasks.

```
In [1]: # TODO: Install any packages that you might need
# For instance, you will need the smdebug package
!pip install smdebug

Collecting smdebug
  Downloading smdebug-1.0.34-py2.py3-none-any.whl.metadata (1.8 kB)
Collecting protobuf<=3.20.3,>=3.20.0 (from smdebug)
  Downloading protobuf-3.20.3-cp310-cp310-manylinux2_12_x86_64.manylinux2010_x86_64.whl.metadata (679 bytes)
Requirement already satisfied: numpy>=1.16.0 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from smdebug) (1.26.4)
Requirement already satisfied: packaging in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from smdebug) (21.3)
Requirement already satisfied: boto3>=1.10.32 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from smdebug) (1.34.142)
Collecting pyinstrument==3.4.2 (from smdebug)
  Downloading pyinstrument-3.4.2-py2.py3-none-any.whl.metadata (22 kB)
Collecting pyinstrument-cext>=0.2.2 (from pyinstrument==3.4.2->smdebug)
  Downloading pyinstrument_cext-0.2.4.tar.gz (4.8 kB)
  Preparing metadata (setup.py) ... done
Requirement already satisfied: botocore<1.35.0,>=1.34.142 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from m boto3>=1.10.32->smdebug) (1.34.142)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from boto3>=1.10.32->smdebug) (1.0.1)
Requirement already satisfied: s3transfer<0.11.0,>=0.10.0 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from m boto3>=1.10.32->smdebug) (0.10.1)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from packaging->smdebug) (3.1.2)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from om botocore<1.35.0,>=1.34.142->boto3>=1.10.32->smdebug) (2.9.0)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from m botocore<1.35.0,>=1.34.142->boto3>=1.10.32->smdebug) (2.2.1)
Requirement already satisfied: six>=1.5 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from python-dateutil<3.0.0,>=2.1->botocore<1.35.0,>=1.34.142->boto3>=1.10.32->smdebug) (1.16.0)
Downloading smdebug-1.0.34-py2.py3-none-any.whl (280 kB)
    |-----| 280.1/280.1 kB 1.3 MB/s eta 0:00:00:00:01
Downloading pyinstrument-3.4.2-py2.py3-none-any.whl (83 kB)
    |-----| 83.3/83.3 kB 8.6 MB/s eta 0:00:00
Downloading protobuf-3.20.3-cp310-cp310-manylinux2_12_x86_64.manylinux2010_x86_64.whl (1.1 MB)
    |-----| 1.1/1.1 MB 50.6 MB/s eta 0:00:00
Building wheels for collected packages: pyinstrument-cext
  Building wheel for pyinstrument-cext (setup.py) ... done
  Created wheel for pyinstrument-cext: filename=pyinstrument_cext-0.2.4-cp310-cp310-linux_x86_64.whl size=6294 sha256=b900dab6e22d8de74f247d64dd7711e93b19e89fb18cb0192fa6d4a63e61eb27
  Stored in directory: /home/ec2-user/.cache/pip/wheels/0f/8b/7a/5f7fd1dd6d3cbb3d350d4c832c5e2f962687749f6d67d573a6
Successfully built pyinstrument-cext
Installing collected packages: pyinstrument-cext, pyinstrument, protobuf, smdebug
  Attempting uninstall: protobuf
    Found existing installation: protobuf 4.25.3
    Uninstalling protobuf-4.25.3:
      Successfully uninstalled protobuf-4.25.3
  Successfully installed protobuf-3.20.3 pyinstrument-3.4.2 pyinstrument-cext-0.2.4 smdebug-1.0.34

In [2]: !pip install --upgrade bokeh==2.4.3
```

```

Collecting bokeh==2.4.3
  Downloading bokeh-2.4.3-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: Jinja2>=2.9 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from bokeh==2.4.3) (3.1.4)
Requirement already satisfied: numpy>=1.11.3 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from bokeh==2.4.3) (1.26.4)
Requirement already satisfied: packaging>=16.8 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from bokeh==2.4.3) (21.3)
Requirement already satisfied: pillow>=7.1.0 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from bokeh==2.4.3) (10.3.0)
Requirement already satisfied: PyYAML>=3.10 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from bokeh==2.4.3) (6.0.1)
Requirement already satisfied: tornado>=5.1 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from bokeh==2.4.3) (6.4)
Requirement already satisfied: typing-extensions>=3.10.0 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from bokeh==2.4.3) (4.12.1)
Requirement already satisfied: MarkupSafe>=2.0 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from Jinja2>=2.9->bokeh==2.4.3) (2.1.5)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in /home/ec2-user/anaconda3/envs/pytorch_p310/lib/python3.10/site-packages (from packaging>=16.8->bokeh==2.4.3) (3.1.2)
Downloading bokeh-2.4.3-py3-none-any.whl (18.5 MB)
18.5/18.5 MB 72.3 MB/s eta 0:00:00:00:0100:01
Installing collected packages: bokeh
  Attempting uninstall: bokeh
    Found existing installation: bokeh 3.4.1
    Uninstalling bokeh-3.4.1:
      Successfully uninstalled bokeh-3.4.1
Successfully installed bokeh-2.4.3

```

```

In [3]: # TODO: Import any packages that you might need
# For instance you will need Boto3 and Sagemaker
import sagemaker
import boto3
from sagemaker.tuner import (
    IntegerParameter,
    CategoricalParameter,
    ContinuousParameter,
    HyperparameterTuner,
)

sagemaker_session = sagemaker.Session()

bucket = sagemaker_session.default_bucket()
prefix = "project/data"

role = sagemaker.get_execution_role()

sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml

```

Dataset

TODO: Explain what dataset you are using for this project. Maybe even give a small overview of the classes, class distributions etc that can help anyone not familiar with the dataset get a better understand of it.

```

In [3]: #TODO: Fetch and upload the data to AWS S3

# Command to download and unzip data
!wget https://s3-us-west-1.amazonaws.com/udacity-aind/dog-project/dogImages.zip
!unzip dogImages.zip

```

```
--2024-08-13 18:45:05-- https://s3-us-west-1.amazonaws.com/udacity-aind/dog-project/dogImages.zip
Resolving s3-us-west-1.amazonaws.com (s3-us-west-1.amazonaws.com)... 52.219.194.80, 52.219.113.152, 52.219.120.176, ...
Connecting to s3-us-west-1.amazonaws.com (s3-us-west-1.amazonaws.com)|52.219.194.80|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1132023110 (1.1G) [application/zip]
Saving to: 'dogImages.zip'
```

```
100%[=====>] 1,132,023,110 43.6MB/s in 40s
```

```
2024-08-13 18:45:45 (27.2 MB/s) - 'dogImages.zip' saved [1132023110/1132023110]
```

```
Archive: dogImages.zip
  creating: dogImages/
    creating: dogImages/test/
      creating: dogImages/train/
        creating: dogImages/valid/
          creating: dogImages/test/001.Affenpinscher/
            inflating: dogImages/test/001.Affenpinscher/Affenpinscher_00003.jpg
            inflating: dogImages/test/001.Affenpinscher/Affenpinscher_00023.jpg
            inflating: dogImages/test/001.Affenpinscher/Affenpinscher_00036.jpg
            inflating: dogImages/test/001.Affenpinscher/Affenpinscher_00047.jpg
            inflating: dogImages/test/001.Affenpinscher/Affenpinscher_00048.jpg
            inflating: dogImages/test/001.Affenpinscher/Affenpinscher_00058.jpg
            inflating: dogImages/test/001.Affenpinscher/Affenpinscher_00071.jpg
            inflating: dogImages/test/001.Affenpinscher/Affenpinscher_00078.jpg
          creating: dogImages/test/002.Afghan_hound/
            inflating: dogImages/test/002.Afghan_hound/Afghan_hound_00116.jpg
            inflating: dogImages/test/002.Afghan_hound/Afghan_hound_00125.jpg
            inflating: dogImages/test/002.Afghan_hound/Afghan_hound_00139.jpg
            inflating: dogImages/test/002.Afghan_hound/Afghan_hound_00141.jpg
            inflating: dogImages/test/002.Afghan_hound/Afghan_hound_00143.jpg
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            inflating: dogImages/test/002.Afghan_hound/Afghan_hound_00149.jpg
            inflating: dogImages/test/002.Afghan_hound/Afghan_hound_00151.jpg
          creating: dogImages/test/003.Airedale_terrier/
            inflating: dogImages/test/003.Airedale_terrier/Airedale_terrier_00166.jpg
            inflating: dogImages/test/003.Airedale_terrier/Airedale_terrier_00175.jpg
            inflating: dogImages/test/003.Airedale_terrier/Airedale_terrier_00179.jpg
            inflating: dogImages/test/003.Airedale_terrier/Airedale_terrier_00207.jpg
            inflating: dogImages/test/003.Airedale_terrier/Airedale_terrier_00210.jpg
            inflating: dogImages/test/003.Airedale_terrier/Airedale_terrier_00214.jpg
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            inflating: dogImages/test/004.Akita/Akita_00258.jpg
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            inflating: dogImages/test/004.Akita/Akita_00263.jpg
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            inflating: dogImages/test/004.Akita/Akita_00276.jpg
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            inflating: dogImages/test/006.American_eskimo_dog/American_eskimo_dog_00452.jpg
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            inflating: dogImages/test/007.American_foxhound/American_foxhound_00522.jpg
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48/145

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89/145

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```

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inflating: dogImages/valid/133.Yorkshire_terrier/Yorkshire_terrier_08348.jpg

```

```

In [19... inputs = sagemaker_session.upload_data(path="dogImages", bucket=bucket, key_prefix=prefix)
print("input spec (in this case, just an S3 path): {}".format(inputs))

input spec (in this case, just an S3 path): s3://sagemaker-us-east-1-065326103865/project/data

```

```

In [4]: inputs = f's3://{bucket}/{prefix}/'
print(inputs)

s3://sagemaker-us-east-1-065326103865/project/data/

```

Hyperparameter Tuning

TODO: This is the part where you will finetune a pretrained model with hyperparameter tuning. Remember that you have to tune a minimum of two hyperparameters. However you are encouraged to tune more. You are also encouraged to explain why you chose to tune those particular hyperparameters and the ranges.

Note: You will need to use the `hpo.py` script to perform hyperparameter tuning.

```

In [3]: #TODO: Declare your HP ranges, metrics etc.
hyperparameter_ranges = {
    "lr": ContinuousParameter(0.001, 0.1),
    "epochs": IntegerParameter(1, 4),
    "batch-size": CategoricalParameter([32, 64, 128, 256]),
}

```

```

In [4]: from sagemaker.pytorch import PyTorch
#TODO: Create estimators for your HPs

# TODO: Your estimator here

estimator = PyTorch(
    entry_point="hpo.py",
    role=role,
    py_version='py36',
    framework_version="1.8",
    instance_count=1,
    instance_type="ml.m5.xlarge"
)

```

```
In [5]: # TODO: Your HP tuner here
objective_metric_name = "Test Loss"
objective_type = "Minimize"
metric_definitions = [{"Name": "Test Loss", "Regex": "Testing loss: ([0-9\\.]+)"}]

tuner = HyperparameterTuner(
    estimator,
    objective_metric_name,
    hyperparameter_ranges,
    metric_definitions,
    max_jobs=4,
    max_parallel_jobs=2,
    objective_type=objective_type,
)
```

```
In [ ]: # TODO: Fit your HP Tuner
# TODO: Remember to include your data channels
tuner.fit({"training": inputs})
```

No finished training job found associated with this estimator. Please make sure this estimator is only used for building workflow config
 No finished training job found associated with this estimator. Please make sure this estimator is only used for building workflow config


```
In [8]: # TODO: Get the best estimators and the best HPs
```

```
best_estimator = tuner.best_estimator() #TODO
```

```
#Get the hyperparameters of the best trained model
best_estimator.hyperparameters()
```

```
2024-08-16 21:55:28 Starting - Preparing the instances for training
2024-08-16 21:55:28 Downloading - Downloading the training image
2024-08-16 21:55:28 Training - Training image download completed. Training in progress.
2024-08-16 21:55:28 Uploading - Uploading generated training model
2024-08-16 21:55:28 Completed - Resource reused by training job: pytorch-training-240816-2050-004-90cfbb39
```

```
Out[8]: {'_tuning_objective_metric': 'Test Loss',
'batch-size': '32',
'epochs': '4',
'lr': '0.013351091326965773',
'sagemaker_container_log_level': '20',
'sagemaker_estimator_class_name': 'PyTorch',
'sagemaker_estimator_module': 'sagemaker.pytorch.estimator',
'sagemaker_job_name': 'pytorch-training-2024-08-16-20-50-17-672',
'sagemaker_program': 'hpo.py',
'sagemaker_region': 'us-east-1',
'sagemaker_submit_directory': 's3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-16-20-50-17-672/source/sourcedir.tar.gz'}
```

Model Profiling and Debugging

TODO: Using the best hyperparameters, create and finetune a new model

Note: You will need to use the `train_model.py` script to perform model profiling and debugging.

```
In [32... from sagemaker.debugger import Rule, ProfilerRule, rule_configs, ProfilerConfig, FrameworkProfile, DebuggerHookConfig, CollectionConfig
```

```
# TODO: Set up debugging and profiling rules and hooks
rules = [
    Rule.sagemaker(rule_configs.vanishing_gradient()),
    Rule.sagemaker(rule_configs.overfit()),
    Rule.sagemaker(rule_configs.overtraining()),
    Rule.sagemaker(rule_configs.poor_weight_initialization()),
    Rule.sagemaker(rule_configs.loss_not_decreasing()),
    ProfilerRule.sagemaker(rule_configs.LowGPUUtilization()),
    ProfilerRule.sagemaker(rule_configs.ProfilerReport())
]

collection_configs = [
    CollectionConfig(
        name="custom_losses",
        parameters={
            "include_regex": "CrossEntropyLoss_output_0",
            "train.save_interval": "100",
            "eval.save_interval": "5"
        }
    )
]
```

```
In [35... debugger_config = DebuggerHookConfig(collection_configs=collection_configs)
```

```
In [36... profiler_config = ProfilerConfig(
    system_monitor_interval_millis=500, framework_profile_params=FrameworkProfile(num_steps=10)
)
```

WARNING:sagemaker.deprecations:Framework profiling will be deprecated from tensorflow 2.12 and pytorch 2.0 in sagemaker>=2.
See: <https://sagemaker.readthedocs.io/en/stable/v2.html> for details.

```
In [37... hyperparameters = {
    "batch-size": 32,
    "epoch": 4,
    "lr": 0.01335,
}
```

```
In [38... from sagemaker.pytorch import PyTorch
# TODO: Create and fit an estimator

# TODO: Your estimator here

estimator = PyTorch(
    role=sagemaker.get_execution_role(),
    instance_count=1,
    instance_type="ml.m5.xlarge",
    entry_point="train_model.py",
    framework_version="1.8",
    py_version="py36",
    hyperparameters=hyperparameters,
    profiler_config=profiler_config,
    rules=rules,
    debugger_hook_config=debugger_config,
)
```

```
In [39... estimator.fit({"training": inputs})
```

INFO:sagemaker.image_uris:image_uri is not presented, retrieving image_uri based on instance_type, framework etc.
 INFO:sagemaker.image_uris:Defaulting to the only supported framework/algorithm version: latest.
 INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
 INFO:sagemaker.image_uris:Defaulting to the only supported framework/algorithm version: latest.
 INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
 INFO:sagemaker.image_uris:Defaulting to the only supported framework/algorithm version: latest.
 INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
 INFO:sagemaker.image_uris:Defaulting to the only supported framework/algorithm version: latest.
 INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
 INFO:sagemaker.image_uris:Defaulting to the only supported framework/algorithm version: latest.
 INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
 INFO:sagemaker.image_uris:Defaulting to the only supported framework/algorithm version: latest.
 INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
 INFO:sagemaker.image_uris:Defaulting to the only supported framework/algorithm version: latest.
 INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.
 INFO:sagemaker.image_uris:image_uri is not presented, retrieving image_uri based on instance_type, framework etc.
 INFO:sagemaker:Creating training-job with name: pytorch-training-2024-08-17-08-45-48-485

```

2024-08-17 08:45:49 Starting - Starting the training job...
2024-08-17 08:46:16 Starting - Preparing the instances for trainingVanishingGradient: InProgress
Overfit: InProgress
Overtraining: InProgress
PoorWeightInitialization: InProgress
LossNotDecreasing: InProgress
LowGPUUtilization: InProgress
ProfilerReport: InProgress
...
2024-08-17 08:46:41 Downloading - Downloading input data.....
2024-08-17 08:47:41 Downloading - Downloading the training image...
2024-08-17 08:48:17 Training - Training image download completed. Training in progress.bash: cannot set terminal process group (-1): Inap
propriate ioctl for device
bash: no job control in this shell
2024-08-17 08:48:02,126 sagemaker-training-toolkit INFO Imported framework sagemaker_pytorch_container.training
2024-08-17 08:48:02,129 sagemaker-training-toolkit INFO No GPUs detected (normal if no gpus installed)
2024-08-17 08:48:02,140 sagemaker_pytorch_container.training INFO Block until all host DNS lookups succeed.
2024-08-17 08:48:02,142 sagemaker_pytorch_container.training INFO Invoking user training script.
2024-08-17 08:48:02,325 sagemaker-training-toolkit INFO No GPUs detected (normal if no gpus installed)
2024-08-17 08:48:02,341 sagemaker-training-toolkit INFO No GPUs detected (normal if no gpus installed)
2024-08-17 08:48:02,354 sagemaker-training-toolkit INFO No GPUs detected (normal if no gpus installed)
2024-08-17 08:48:02,364 sagemaker-training-toolkit INFO Invoking user script
Training Env:
{
  "additional_framework_parameters": {},
  "channel_input_dirs": {
    "training": "/opt/ml/input/data/training"
  },
  "current_host": "algo-1",
  "framework_module": "sagemaker_pytorch_container.training:main",
  "hosts": [
    "algo-1"
  ],
  "hyperparameters": {
    "batch-size": 32,
    "epoch": 4,
    "lr": 0.01335
  },
  "input_config_dir": "/opt/ml/input/config",
  "input_data_config": {
    "training": {
      "TrainingInputMode": "File",
      "S3DistributionType": "FullyReplicated",
      "RecordWrapperType": "None"
    }
  },
  "input_dir": "/opt/ml/input",
  "is_master": true,
  "job_name": "pytorch-training-2024-08-17-08-45-48-485",
  "log_level": 20,
  "master_hostname": "algo-1",
  "model_dir": "/opt/ml/model",
  "module_dir": "s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/source/sourcedir.tar.gz",
  "module_name": "train_model",
  "network_interface_name": "eth0",
  "num_cpus": 4,
  "num_gpus": 0,
  "output_data_dir": "/opt/ml/output/data",
  "output_dir": "/opt/ml/output",
  "output_intermediate_dir": "/opt/ml/output/intermediate",
  "resource_config": {
    "current_host": "algo-1",
    "current_instance_type": "ml.m5.xlarge",
    "current_group_name": "homogeneousCluster",
    "hosts": [
      "algo-1"
    ],
    "instance_groups": [
      {
        "instance_group_name": "homogeneousCluster",
        "instance_type": "ml.m5.xlarge",
        "hosts": [
          "algo-1"
        ]
      }
    ],
    "network_interface_name": "eth0"
  },
  "user_entry_point": "train_model.py"
}
Environment variables:
SM_HOSTS=["algo-1"]
SM_NETWORK_INTERFACE_NAME=eth0
SM_HPS={"batch-size":32,"epoch":4,"lr":0.01335}

```

```

SM_USER_ENTRY_POINT=train_model.py
SM_FRAMEWORK_PARAMS={}
SM_RESOURCE_CONFIG={"current_group_name":"homogeneousCluster","current_host":"algo-1","current_instance_type":"ml.m5.xlarge","hosts":["algo-1"],"instance_groups":[{"hosts":["algo-1"],"instance_group_name":"homogeneousCluster","instance_type":"ml.m5.xlarge"}],"network_interface_name":"eth0"}
SM_INPUT_DATA_CONFIG={"training":{"RecordWrapperType":"None","S3DistributionType":"FullyReplicated","TrainingInputMode":"File"}}
SM_OUTPUT_DATA_DIR=/opt/ml/output/data
SM_CHANNELS=["training"]
SM_CURRENT_HOST=algo-1
SM_MODULE_NAME=train_model
SM_LOG_LEVEL=20
SM_FRAMEWORK_MODULE=sagemaker_pytorch_container.training:main
SM_INPUT_DIR=/opt/ml/input
SM_INPUT_CONFIG_DIR=/opt/ml/input/config
SM_OUTPUT_DIR=/opt/ml/output
SM_NUM_CPUS=4
SM_NUM_GPUS=0
SM_MODEL_DIR=/opt/ml/model
SM_MODULE_DIR=s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/source/sourcedir.tar.gz
SM_TRAINING_ENV={"additional_framework_parameters":{},"channel_input_dirs":{"training":"/opt/ml/input/data/training"},"current_host":"algo-1","framework_module":"sagemaker_pytorch_container.training:main","hosts":["algo-1"],"hyperparameters":{"batch-size":32,"epoch":4,"lr":0.01335},"input_config_dir":"/opt/ml/input/config","input_data_config":{"training":{"RecordWrapperType":"None","S3DistributionType":"FullyReplicated","TrainingInputMode":"File"},"input_dir":"/opt/ml/input","is_master":true,"job_name":"pytorch-training-2024-08-17-08-45-48-485","log_level":20,"master_hostname":"algo-1","model_dir":"/opt/ml/model","module_dir":"s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/source/sourcedir.tar.gz","module_name":"train_model","network_interface_name":"eth0","num_cpus":4,"num_gpus":0,"output_data_dir":"/opt/ml/output/data","output_dir":"/opt/ml/output","output_intermediate_dir":"/opt/ml/output/intermediate"},"resource_config":{"current_group_name":"homogeneousCluster","current_host":"algo-1","current_instance_type":"ml.m5.xlarge","hosts":["algo-1"],"instance_groups":[{"hosts":["algo-1"],"instance_group_name":"homogeneousCluster","instance_type":"ml.m5.xlarge"}],"network_interface_name":"eth0"},"user_entry_point":"train_model.py"}
SM_USER_ARGS=["--batch-size","32","--epoch","4","--lr","0.01335"]
SM_OUTPUT_INTERMEDIATE_DIR=/opt/ml/output/intermediate
SM_CHANNEL_TRAINING=/opt/ml/input/data/training
SM_HP_BATCH_SIZE=32
SM_HP_EPOCH=4
SM_HP_LR=0.01335
PYTHONPATH=/opt/ml/code:/opt/conda/bin:/opt/conda/lib/python3.6:/opt/conda/lib/python3.6:/opt/conda/lib/python3.6/lib-dynload:/opt/conda/lib/python3.6/site-packages
Invoking script with the following command:
/opt/conda/bin/python3.6 train_model.py --batch-size 32 --epoch 4 --lr 0.01335
[2024-08-17 08:48:03.208 algo-1:28 INFO utils.py:27] RULE_JOB_STOP_SIGNAL_FILE_NAME: None
[2024-08-17 08:48:03.433 algo-1:28 INFO profiler_config_parser.py:102] Using config at /opt/ml/input/config/profilerconfig.json.
[2024-08-17 08:48:04.439 algo-1:28 INFO json_config.py:91] Creating hook from json_config at /opt/ml/input/config/debughookconfig.json.
[2024-08-17 08:48:04.441 algo-1:28 INFO hook.py:201] tensorboard_dir has not been set for the hook. SMDebug will not be exporting tensorboard summaries.
[2024-08-17 08:48:04.442 algo-1:28 INFO hook.py:255] Saving to /opt/ml/output/tensors
[2024-08-17 08:48:04.442 algo-1:28 INFO state_store.py:77] The checkpoint config file /opt/ml/input/config/checkpointconfig.json does not exist.
[2024-08-17 08:48:04.457 algo-1:28 INFO hook.py:591] name:fc.0.weight count_params:131072
[2024-08-17 08:48:04.457 algo-1:28 INFO hook.py:591] name:fc.0.bias count_params:256
[2024-08-17 08:48:04.457 algo-1:28 INFO hook.py:591] name:fc.2.weight count_params:34048
[2024-08-17 08:48:06.921 algo-1:28 INFO hook.py:591] name:fc.2.bias count_params:133
[2024-08-17 08:48:04.458 algo-1:28 INFO hook.py:593] Total Trainable Params: 165509
start training
[2024-08-17 08:48:05.188 algo-1:28 INFO hook.py:425] Monitoring the collections: gradients, losses, relu_input, custom_losses
[2024-08-17 08:48:05.190 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/prestepzero.*-start-1723884483433523.0_train-0-stepstart-1723884485189720.0/python_stats.
[2024-08-17 08:48:05.206 algo-1:28 INFO hook.py:488] Hook is writing from the hook with pid: 28
[2024-08-17 08:48:06.921 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-0-stepstart-1723884485200132.5_train-0-forwardpassend-1723884486921327.0/python_stats.
Train Epoch: 1 [32/6680 (0%)] Loss: 0.0233
[2024-08-17 08:48:07.326 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-0-forwardpassend-1723884486924864.2_train-1-stepstart-1723884487326307.5/python_stats.
[2024-08-17 08:48:12.807 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-1-stepstart-1723884487330826.2_train-1-forwardpassend-1723884492806800.8/python_stats.
Train Epoch: 1 [64/6680 (1%)] Loss: 0.0729
[2024-08-17 08:48:13.232 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-1-forwardpassend-1723884492809263.5_train-2-stepstart-1723884493232415.5/python_stats.
[2024-08-17 08:48:18.036 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-2-stepstart-1723884493235821.2_train-2-forwardpassend-1723884498035816.0/python_stats.
Train Epoch: 1 [96/6680 (1%)] Loss: 0.1024
[2024-08-17 08:48:18.367 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-2-forwardpassend-1723884498037632.0_train-3-stepstart-1723884498366988.0/python_stats.
[2024-08-17 08:48:23.263 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-3-stepstart-1723884498369875.2_train-3-forwardpassend-1723884503262922.0/python_stats.
Train Epoch: 1 [128/6680 (2%)] Loss: 0.1314
[2024-08-17 08:48:23.727 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-3-forwardpassend-1723884503264664.5_train-4-stepstart-1723884503726533.8/python_stats.
[2024-08-17 08:48:28.598 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-4-stepstart-1723884503730322.0_train-4-forwardpassend-1723884508598291.8/python_stats.
Train Epoch: 1 [160/6680 (2%)] Loss: 0.1548
[2024-08-17 08:48:29.104 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-4-forwardpassend-1723884508600032.5_train-5-stepstart-1723884509103377.0/python_stats.
[2024-08-17 08:48:33.933 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-5-stepstart-1723884509107257.5_train-5-forwardpassend-1723884513932783.8/python_stats.

```

```
Train Epoch: 1 [192/6680 (3%)] Loss: 0.1790
[2024-08-17 08:48:34.255 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-5-forwardpassend-1723884513934723.8_train-6-stepstart-1723884514254861.5/python_stats.
[2024-08-17 08:48:39.140 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-6-stepstart-1723884514258113.5_train-6-forwardpassend-1723884519140458.8/python_stats.
Train Epoch: 1 [224/6680 (3%)] Loss: 0.2027
[2024-08-17 08:48:39.446 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-6-forwardpassend-1723884519142300.2_train-7-stepstart-1723884519445982.8/python_stats.
[2024-08-17 08:48:44.345 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-7-stepstart-1723884519449449.0_train-7-forwardpassend-1723884524345290.0/python_stats.
Train Epoch: 1 [256/6680 (4%)] Loss: 0.2259
[2024-08-17 08:48:44.926 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-7-forwardpassend-1723884524347104.8_train-8-stepstart-1723884524926149.2/python_stats.
[2024-08-17 08:48:49.705 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-8-stepstart-1723884524929691.5_train-8-forwardpassend-1723884529704792.8/python_stats.
Train Epoch: 1 [288/6680 (4%)] Loss: 0.2486
[2024-08-17 08:48:50.180 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-8-forwardpassend-1723884529706559.8_train-9-stepstart-1723884530180189.0/python_stats.
[2024-08-17 08:48:55.021 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-9-stepstart-1723884530183474.8_train-9-forwardpassend-1723884535021432.8/python_stats.
Train Epoch: 1 [320/6680 (5%)] Loss: 0.2716
[2024-08-17 08:48:55.529 algo-1:28 INFO python_profiler.py:182] Dumping cProfile stats to /opt/ml/output/profiler/framework/pytorch/cprofile/28-algo-1/train-9-forwardpassend-1723884535023454.5_train-10-stepstart-1723884535528428.2/python_stats.
Train Epoch: 1 [352/6680 (5%)] Loss: 0.2946
Train Epoch: 1 [384/6680 (6%)] Loss: 0.3178
Train Epoch: 1 [416/6680 (6%)] Loss: 0.3411
Train Epoch: 1 [448/6680 (7%)] Loss: 0.3640
Train Epoch: 1 [480/6680 (7%)] Loss: 0.3871
Train Epoch: 1 [512/6680 (8%)] Loss: 0.4105
Train Epoch: 1 [544/6680 (8%)] Loss: 0.4332
Train Epoch: 1 [576/6680 (9%)] Loss: 0.4559
Train Epoch: 1 [608/6680 (9%)] Loss: 0.4787
Train Epoch: 1 [640/6680 (10%)] Loss: 0.5016
Train Epoch: 1 [672/6680 (10%)] Loss: 0.5250
Train Epoch: 1 [704/6680 (11%)] Loss: 0.5474
Train Epoch: 1 [736/6680 (11%)] Loss: 0.5706
Train Epoch: 1 [768/6680 (11%)] Loss: 0.5932
Train Epoch: 1 [800/6680 (12%)] Loss: 0.6157
Train Epoch: 1 [832/6680 (12%)] Loss: 0.6387
Train Epoch: 1 [864/6680 (13%)] Loss: 0.6602
Train Epoch: 1 [896/6680 (13%)] Loss: 0.6816
Train Epoch: 1 [928/6680 (14%)] Loss: 0.7036
Train Epoch: 1 [960/6680 (14%)] Loss: 0.7273
Train Epoch: 1 [992/6680 (15%)] Loss: 0.7493
Train Epoch: 1 [1024/6680 (15%)] Loss: 0.7716
Train Epoch: 1 [1056/6680 (16%)] Loss: 0.7913
Train Epoch: 1 [1088/6680 (16%)] Loss: 0.8133
Train Epoch: 1 [1120/6680 (17%)] Loss: 0.8341
Train Epoch: 1 [1152/6680 (17%)] Loss: 0.8552
Train Epoch: 1 [1184/6680 (18%)] Loss: 0.8763
Train Epoch: 1 [1216/6680 (18%)] Loss: 0.8987
Train Epoch: 1 [1248/6680 (19%)] Loss: 0.9200
Train Epoch: 1 [1280/6680 (19%)] Loss: 0.9410
Train Epoch: 1 [1312/6680 (20%)] Loss: 0.9623
Train Epoch: 1 [1344/6680 (20%)] Loss: 0.9838
Train Epoch: 1 [1376/6680 (21%)] Loss: 1.0048
Train Epoch: 1 [1408/6680 (21%)] Loss: 1.0260
Train Epoch: 1 [1440/6680 (22%)] Loss: 1.0474
Train Epoch: 1 [1472/6680 (22%)] Loss: 1.0669
Train Epoch: 1 [1504/6680 (23%)] Loss: 1.0876
Train Epoch: 1 [1536/6680 (23%)] Loss: 1.1078
Train Epoch: 1 [1568/6680 (23%)] Loss: 1.1271
Train Epoch: 1 [1600/6680 (24%)] Loss: 1.1469
Train Epoch: 1 [1632/6680 (24%)] Loss: 1.1668
Train Epoch: 1 [1664/6680 (25%)] Loss: 1.1870
Train Epoch: 1 [1696/6680 (25%)] Loss: 1.2070
Train Epoch: 1 [1728/6680 (26%)] Loss: 1.2275
Train Epoch: 1 [1760/6680 (26%)] Loss: 1.2461
Train Epoch: 1 [1792/6680 (27%)] Loss: 1.2659
Train Epoch: 1 [1824/6680 (27%)] Loss: 1.2852
Train Epoch: 1 [1856/6680 (28%)] Loss: 1.3041
Train Epoch: 1 [1888/6680 (28%)] Loss: 1.3234
Train Epoch: 1 [1920/6680 (29%)] Loss: 1.3429
Train Epoch: 1 [1952/6680 (29%)] Loss: 1.3628
Train Epoch: 1 [1984/6680 (30%)] Loss: 1.3823
Train Epoch: 1 [2016/6680 (30%)] Loss: 1.4006
Train Epoch: 1 [2048/6680 (31%)] Loss: 1.4193
Train Epoch: 1 [2080/6680 (31%)] Loss: 1.4372
Train Epoch: 1 [2112/6680 (32%)] Loss: 1.4549
Train Epoch: 1 [2144/6680 (32%)] Loss: 1.4739
Train Epoch: 1 [2176/6680 (33%)] Loss: 1.4921
Train Epoch: 1 [2208/6680 (33%)] Loss: 1.5103
Train Epoch: 1 [2240/6680 (34%)] Loss: 1.5273
Train Epoch: 1 [2272/6680 (34%)] Loss: 1.5456
```


Train Epoch: 1 [2304/6680 (34%)] Loss: 1.5640
Train Epoch: 1 [2336/6680 (35%)] Loss: 1.5806
Train Epoch: 1 [2368/6680 (35%)] Loss: 1.5958
Train Epoch: 1 [2400/6680 (36%)] Loss: 1.6139
Train Epoch: 1 [2432/6680 (36%)] Loss: 1.6303
Train Epoch: 1 [2464/6680 (37%)] Loss: 1.6465
Train Epoch: 1 [2496/6680 (37%)] Loss: 1.6621
Train Epoch: 1 [2528/6680 (38%)] Loss: 1.6798
Train Epoch: 1 [2560/6680 (38%)] Loss: 1.6969
Train Epoch: 1 [2592/6680 (39%)] Loss: 1.7132
Train Epoch: 1 [2624/6680 (39%)] Loss: 1.7301
Train Epoch: 1 [2656/6680 (40%)] Loss: 1.7469
Train Epoch: 1 [2688/6680 (40%)] Loss: 1.7635
Train Epoch: 1 [2720/6680 (41%)] Loss: 1.7790
Train Epoch: 1 [2752/6680 (41%)] Loss: 1.7975
Train Epoch: 1 [2784/6680 (42%)] Loss: 1.8142
Train Epoch: 1 [2816/6680 (42%)] Loss: 1.8290
Train Epoch: 1 [2848/6680 (43%)] Loss: 1.8450
Train Epoch: 1 [2880/6680 (43%)] Loss: 1.8596
Train Epoch: 1 [2912/6680 (44%)] Loss: 1.8752
Train Epoch: 1 [2944/6680 (44%)] Loss: 1.8904
Train Epoch: 1 [2976/6680 (45%)] Loss: 1.9038
Train Epoch: 1 [3008/6680 (45%)] Loss: 1.9185
Train Epoch: 1 [3040/6680 (46%)] Loss: 1.9325
Train Epoch: 1 [3072/6680 (46%)] Loss: 1.9463
Train Epoch: 1 [3104/6680 (46%)] Loss: 1.9613
Train Epoch: 1 [3136/6680 (47%)] Loss: 1.9751
Train Epoch: 1 [3168/6680 (47%)] Loss: 1.9889
Train Epoch: 1 [3200/6680 (48%)] Loss: 2.0030
Train Epoch: 1 [3232/6680 (48%)] Loss: 2.0172
Train Epoch: 1 [3264/6680 (49%)] Loss: 2.0319
Train Epoch: 1 [3296/6680 (49%)] Loss: 2.0440
Train Epoch: 1 [3328/6680 (50%)] Loss: 2.0586
Train Epoch: 1 [3360/6680 (50%)] Loss: 2.0742
Train Epoch: 1 [3392/6680 (51%)] Loss: 2.0872
Train Epoch: 1 [3424/6680 (51%)] Loss: 2.1015
Train Epoch: 1 [3456/6680 (52%)] Loss: 2.1146
Train Epoch: 1 [3488/6680 (52%)] Loss: 2.1260
Train Epoch: 1 [3520/6680 (53%)] Loss: 2.1383
Train Epoch: 1 [3552/6680 (53%)] Loss: 2.1512
Train Epoch: 1 [3584/6680 (54%)] Loss: 2.1637
Train Epoch: 1 [3616/6680 (54%)] Loss: 2.1767
Train Epoch: 1 [3648/6680 (55%)] Loss: 2.1901
Train Epoch: 1 [3680/6680 (55%)] Loss: 2.2033
Train Epoch: 1 [3712/6680 (56%)] Loss: 2.2161
VanishingGradient: InProgress
Overfit: InProgress
Overtraining: InProgress
PoorWeightInitialization: IssuesFound
LossNotDecreasing: InProgress
Train Epoch: 1 [3744/6680 (56%)] Loss: 2.2274
Train Epoch: 1 [3776/6680 (57%)] Loss: 2.2412
Train Epoch: 1 [3808/6680 (57%)] Loss: 2.2527
Train Epoch: 1 [3840/6680 (57%)] Loss: 2.2658
Train Epoch: 1 [3872/6680 (58%)] Loss: 2.2790
Train Epoch: 1 [3904/6680 (58%)] Loss: 2.2933
Train Epoch: 1 [3936/6680 (59%)] Loss: 2.3060
Train Epoch: 1 [3968/6680 (59%)] Loss: 2.3183
Train Epoch: 1 [4000/6680 (60%)] Loss: 2.3312
Train Epoch: 1 [4032/6680 (60%)] Loss: 2.3423
Train Epoch: 1 [4064/6680 (61%)] Loss: 2.3549
Train Epoch: 1 [4096/6680 (61%)] Loss: 2.3678
Train Epoch: 1 [4128/6680 (62%)] Loss: 2.3816
Train Epoch: 1 [4160/6680 (62%)] Loss: 2.3920
Train Epoch: 1 [4192/6680 (63%)] Loss: 2.4040
Train Epoch: 1 [4224/6680 (63%)] Loss: 2.4145
Train Epoch: 1 [4256/6680 (64%)] Loss: 2.4252
Train Epoch: 1 [4288/6680 (64%)] Loss: 2.4365
Train Epoch: 1 [4320/6680 (65%)] Loss: 2.4478
Train Epoch: 1 [4352/6680 (65%)] Loss: 2.4586
Train Epoch: 1 [4384/6680 (66%)] Loss: 2.4707
Train Epoch: 1 [4416/6680 (66%)] Loss: 2.4820
Train Epoch: 1 [4448/6680 (67%)] Loss: 2.4933
Train Epoch: 1 [4480/6680 (67%)] Loss: 2.5036
Train Epoch: 1 [4512/6680 (68%)] Loss: 2.5137
Train Epoch: 1 [4544/6680 (68%)] Loss: 2.5239
Train Epoch: 1 [4576/6680 (69%)] Loss: 2.5352
Train Epoch: 1 [4608/6680 (69%)] Loss: 2.5458
Train Epoch: 1 [4640/6680 (69%)] Loss: 2.5569
Train Epoch: 1 [4672/6680 (70%)] Loss: 2.5692
Train Epoch: 1 [4704/6680 (70%)] Loss: 2.5809
Train Epoch: 1 [4736/6680 (71%)] Loss: 2.5914
Train Epoch: 1 [4768/6680 (71%)] Loss: 2.6026
Train Epoch: 1 [4800/6680 (72%)] Loss: 2.6136

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Train Epoch: 1 [4832/6680 (72%)] Loss: 2.6238
Train Epoch: 1 [4864/6680 (73%)] Loss: 2.6355
Train Epoch: 1 [4896/6680 (73%)] Loss: 2.6462
Train Epoch: 1 [4928/6680 (74%)] Loss: 2.6560
Train Epoch: 1 [4960/6680 (74%)] Loss: 2.6661
Train Epoch: 1 [4992/6680 (75%)] Loss: 2.6763
Train Epoch: 1 [5024/6680 (75%)] Loss: 2.6878
Train Epoch: 1 [5056/6680 (76%)] Loss: 2.6974
Train Epoch: 1 [5088/6680 (76%)] Loss: 2.7076
Train Epoch: 1 [5120/6680 (77%)] Loss: 2.7181
Train Epoch: 1 [5152/6680 (77%)] Loss: 2.7287
Train Epoch: 1 [5184/6680 (78%)] Loss: 2.7382
Train Epoch: 1 [5216/6680 (78%)] Loss: 2.7490
Train Epoch: 1 [5248/6680 (79%)] Loss: 2.7594
Train Epoch: 1 [5280/6680 (79%)] Loss: 2.7682
Train Epoch: 1 [5312/6680 (80%)] Loss: 2.7788
Train Epoch: 1 [5344/6680 (80%)] Loss: 2.7874
Train Epoch: 1 [5376/6680 (80%)] Loss: 2.7967
Train Epoch: 1 [5408/6680 (81%)] Loss: 2.8064
Train Epoch: 1 [5440/6680 (81%)] Loss: 2.8161
Train Epoch: 1 [5472/6680 (82%)] Loss: 2.8257
Train Epoch: 1 [5504/6680 (82%)] Loss: 2.8347
Train Epoch: 1 [5536/6680 (83%)] Loss: 2.8433
Train Epoch: 1 [5568/6680 (83%)] Loss: 2.8520
Train Epoch: 1 [5600/6680 (84%)] Loss: 2.8619
Train Epoch: 1 [5632/6680 (84%)] Loss: 2.8706
Train Epoch: 1 [5664/6680 (85%)] Loss: 2.8802
Train Epoch: 1 [5696/6680 (85%)] Loss: 2.8902
Train Epoch: 1 [5728/6680 (86%)] Loss: 2.8998
Train Epoch: 1 [5760/6680 (86%)] Loss: 2.9106
Train Epoch: 1 [5792/6680 (87%)] Loss: 2.9207
Train Epoch: 1 [5824/6680 (87%)] Loss: 2.9297
Train Epoch: 1 [5856/6680 (88%)] Loss: 2.9379
Train Epoch: 1 [5888/6680 (88%)] Loss: 2.9488
Train Epoch: 1 [5920/6680 (89%)] Loss: 2.9593
Train Epoch: 1 [5952/6680 (89%)] Loss: 2.9677
Train Epoch: 1 [5984/6680 (90%)] Loss: 2.9785
Train Epoch: 1 [6016/6680 (90%)] Loss: 2.9884
Train Epoch: 1 [6048/6680 (91%)] Loss: 2.9986
Train Epoch: 1 [6080/6680 (91%)] Loss: 3.0076
Train Epoch: 1 [6112/6680 (91%)] Loss: 3.0170
Train Epoch: 1 [6144/6680 (92%)] Loss: 3.0269
Train Epoch: 1 [6176/6680 (92%)] Loss: 3.0352
Train Epoch: 1 [6208/6680 (93%)] Loss: 3.0439
Train Epoch: 1 [6240/6680 (93%)] Loss: 3.0530
Train Epoch: 1 [6272/6680 (94%)] Loss: 3.0616
Train Epoch: 1 [6304/6680 (94%)] Loss: 3.0715
Train Epoch: 1 [6336/6680 (95%)] Loss: 3.0795
Train Epoch: 1 [6368/6680 (95%)] Loss: 3.0873
Train Epoch: 1 [6400/6680 (96%)] Loss: 3.0950
Train Epoch: 1 [6432/6680 (96%)] Loss: 3.1021
Train Epoch: 1 [6464/6680 (97%)] Loss: 3.1104
Train Epoch: 1 [6496/6680 (97%)] Loss: 3.1197
Train Epoch: 1 [6528/6680 (98%)] Loss: 3.1271
Train Epoch: 1 [6560/6680 (98%)] Loss: 3.1362
Train Epoch: 1 [6592/6680 (99%)] Loss: 3.1438
Train Epoch: 1 [6624/6680 (99%)] Loss: 3.1539
Train Epoch: 1 [6656/6680 (100%)] Loss: 3.1612
Train Epoch: 1 [5016/6680 (75%)] Loss: 3.1707
start validating
Validate Epoch: 1 [32/835 (4%)] Loss: 0.0511
Validate Epoch: 1 [64/835 (8%)] Loss: 0.1129
Validate Epoch: 1 [96/835 (11%)] Loss: 0.1757
Validate Epoch: 1 [128/835 (15%)] Loss: 0.2188
Validate Epoch: 1 [160/835 (19%)] Loss: 0.2462
Validate Epoch: 1 [192/835 (23%)] Loss: 0.2973
Validate Epoch: 1 [224/835 (27%)] Loss: 0.3589
Validate Epoch: 1 [256/835 (31%)] Loss: 0.4376
Validate Epoch: 1 [288/835 (34%)] Loss: 0.4915
Validate Epoch: 1 [320/835 (38%)] Loss: 0.5439
Validate Epoch: 1 [352/835 (42%)] Loss: 0.6155
Validate Epoch: 1 [384/835 (46%)] Loss: 0.6678
Validate Epoch: 1 [416/835 (50%)] Loss: 0.7373
Validate Epoch: 1 [448/835 (54%)] Loss: 0.7922
Validate Epoch: 1 [480/835 (57%)] Loss: 0.8703
Validate Epoch: 1 [512/835 (61%)] Loss: 0.9590
Validate Epoch: 1 [544/835 (65%)] Loss: 1.0294
Validate Epoch: 1 [576/835 (69%)] Loss: 1.0913
Validate Epoch: 1 [608/835 (73%)] Loss: 1.1495
Validate Epoch: 1 [640/835 (77%)] Loss: 1.1979
Validate Epoch: 1 [672/835 (80%)] Loss: 1.2764
Validate Epoch: 1 [704/835 (84%)] Loss: 1.3423
Validate Epoch: 1 [736/835 (88%)] Loss: 1.4149
Validate Epoch: 1 [768/835 (92%)] Loss: 1.4816
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Validate Epoch: 1 [800/835 (96%)] Loss: 1.5521
Validate Epoch: 1 [832/835 (100%)] Loss: 1.6225
Validate Epoch: 1 [81/835 (10%)] Loss: 1.7391
start training
Train Epoch: 2 [32/6680 (0%)] Loss: 0.0071
Train Epoch: 2 [64/6680 (1%)] Loss: 0.0176
Train Epoch: 2 [96/6680 (1%)] Loss: 0.0253
Train Epoch: 2 [128/6680 (2%)] Loss: 0.0357
Train Epoch: 2 [160/6680 (2%)] Loss: 0.0429
Train Epoch: 2 [192/6680 (3%)] Loss: 0.0526
Train Epoch: 2 [224/6680 (3%)] Loss: 0.0614
Train Epoch: 2 [256/6680 (4%)] Loss: 0.0686
Train Epoch: 2 [288/6680 (4%)] Loss: 0.0780
Train Epoch: 2 [320/6680 (5%)] Loss: 0.0864
Train Epoch: 2 [352/6680 (5%)] Loss: 0.0962
Train Epoch: 2 [384/6680 (6%)] Loss: 0.1039
Train Epoch: 2 [416/6680 (6%)] Loss: 0.1125
Train Epoch: 2 [448/6680 (7%)] Loss: 0.1210
Train Epoch: 2 [480/6680 (7%)] Loss: 0.1288
Train Epoch: 2 [512/6680 (8%)] Loss: 0.1375
Train Epoch: 2 [544/6680 (8%)] Loss: 0.1450
Train Epoch: 2 [576/6680 (9%)] Loss: 0.1512
Train Epoch: 2 [608/6680 (9%)] Loss: 0.1583
Train Epoch: 2 [640/6680 (10%)] Loss: 0.1669
Train Epoch: 2 [672/6680 (10%)] Loss: 0.1753
Train Epoch: 2 [704/6680 (11%)] Loss: 0.1824
Train Epoch: 2 [736/6680 (11%)] Loss: 0.1900
Train Epoch: 2 [768/6680 (11%)] Loss: 0.1971
Train Epoch: 2 [800/6680 (12%)] Loss: 0.2054
Train Epoch: 2 [832/6680 (12%)] Loss: 0.2128
Train Epoch: 2 [864/6680 (13%)] Loss: 0.2196
Train Epoch: 2 [896/6680 (13%)] Loss: 0.2280
Train Epoch: 2 [928/6680 (14%)] Loss: 0.2367
Train Epoch: 2 [960/6680 (14%)] Loss: 0.2450
Train Epoch: 2 [992/6680 (15%)] Loss: 0.2524
Train Epoch: 2 [1024/6680 (15%)] Loss: 0.2597
Train Epoch: 2 [1056/6680 (16%)] Loss: 0.2676
Train Epoch: 2 [1088/6680 (16%)] Loss: 0.2746
Train Epoch: 2 [1120/6680 (17%)] Loss: 0.2813
Train Epoch: 2 [1152/6680 (17%)] Loss: 0.2889
Train Epoch: 2 [1184/6680 (18%)] Loss: 0.2965
Train Epoch: 2 [1216/6680 (18%)] Loss: 0.3051
Train Epoch: 2 [1248/6680 (19%)] Loss: 0.3115
Train Epoch: 2 [1280/6680 (19%)] Loss: 0.3197
Train Epoch: 2 [1312/6680 (20%)] Loss: 0.3273
Train Epoch: 2 [1344/6680 (20%)] Loss: 0.3347
Train Epoch: 2 [1376/6680 (21%)] Loss: 0.3435
Train Epoch: 2 [1408/6680 (21%)] Loss: 0.3508
Train Epoch: 2 [1440/6680 (22%)] Loss: 0.3567
Train Epoch: 2 [1472/6680 (22%)] Loss: 0.3643
Train Epoch: 2 [1504/6680 (23%)] Loss: 0.3706
Train Epoch: 2 [1536/6680 (23%)] Loss: 0.3785
Train Epoch: 2 [1568/6680 (23%)] Loss: 0.3853
Train Epoch: 2 [1600/6680 (24%)] Loss: 0.3934
Train Epoch: 2 [1632/6680 (24%)] Loss: 0.4011
Train Epoch: 2 [1664/6680 (25%)] Loss: 0.4071
Train Epoch: 2 [1696/6680 (25%)] Loss: 0.4132
Train Epoch: 2 [1728/6680 (26%)] Loss: 0.4208
Train Epoch: 2 [1760/6680 (26%)] Loss: 0.4282
Train Epoch: 2 [1792/6680 (27%)] Loss: 0.4353
Train Epoch: 2 [1824/6680 (27%)] Loss: 0.4413
Train Epoch: 2 [1856/6680 (28%)] Loss: 0.4486
Train Epoch: 2 [1888/6680 (28%)] Loss: 0.4573
Train Epoch: 2 [1920/6680 (29%)] Loss: 0.4639
Train Epoch: 2 [1952/6680 (29%)] Loss: 0.4714
Train Epoch: 2 [1984/6680 (30%)] Loss: 0.4784
Train Epoch: 2 [2016/6680 (30%)] Loss: 0.4857
Train Epoch: 2 [2048/6680 (31%)] Loss: 0.4921
Train Epoch: 2 [2080/6680 (31%)] Loss: 0.4989
Train Epoch: 2 [2112/6680 (32%)] Loss: 0.5049
Train Epoch: 2 [2144/6680 (32%)] Loss: 0.5121
Train Epoch: 2 [2176/6680 (33%)] Loss: 0.5184
Train Epoch: 2 [2208/6680 (33%)] Loss: 0.5254
Train Epoch: 2 [2240/6680 (34%)] Loss: 0.5328
Train Epoch: 2 [2272/6680 (34%)] Loss: 0.5399
Train Epoch: 2 [2304/6680 (34%)] Loss: 0.5465
Train Epoch: 2 [2336/6680 (35%)] Loss: 0.5525
Train Epoch: 2 [2368/6680 (35%)] Loss: 0.5578
Train Epoch: 2 [2400/6680 (36%)] Loss: 0.5652
Train Epoch: 2 [2432/6680 (36%)] Loss: 0.5710
Train Epoch: 2 [2464/6680 (37%)] Loss: 0.5788
Train Epoch: 2 [2496/6680 (37%)] Loss: 0.5860
Train Epoch: 2 [2528/6680 (38%)] Loss: 0.5930
Train Epoch: 2 [2560/6680 (38%)] Loss: 0.5992
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Train Epoch: 2 [2592/6680 (39%)] Loss: 0.6074
Train Epoch: 2 [2624/6680 (39%)] Loss: 0.6139
Train Epoch: 2 [2656/6680 (40%)] Loss: 0.6192
Train Epoch: 2 [2688/6680 (40%)] Loss: 0.6256
Train Epoch: 2 [2720/6680 (41%)] Loss: 0.6326
Train Epoch: 2 [2752/6680 (41%)] Loss: 0.6401
Train Epoch: 2 [2784/6680 (42%)] Loss: 0.6475
Train Epoch: 2 [2816/6680 (42%)] Loss: 0.6554
Train Epoch: 2 [2848/6680 (43%)] Loss: 0.6609
Train Epoch: 2 [2880/6680 (43%)] Loss: 0.6675
Train Epoch: 2 [2912/6680 (44%)] Loss: 0.6742
Train Epoch: 2 [2944/6680 (44%)] Loss: 0.6789
Train Epoch: 2 [2976/6680 (45%)] Loss: 0.6852
Train Epoch: 2 [3008/6680 (45%)] Loss: 0.6927
Train Epoch: 2 [3040/6680 (46%)] Loss: 0.7000
Train Epoch: 2 [3072/6680 (46%)] Loss: 0.7077
Train Epoch: 2 [3104/6680 (46%)] Loss: 0.7136
Train Epoch: 2 [3136/6680 (47%)] Loss: 0.7194
Train Epoch: 2 [3168/6680 (47%)] Loss: 0.7262
Train Epoch: 2 [3200/6680 (48%)] Loss: 0.7332
Train Epoch: 2 [3232/6680 (48%)] Loss: 0.7395
Train Epoch: 2 [3264/6680 (49%)] Loss: 0.7471
Train Epoch: 2 [3296/6680 (49%)] Loss: 0.7538
Train Epoch: 2 [3328/6680 (50%)] Loss: 0.7605
Train Epoch: 2 [3360/6680 (50%)] Loss: 0.7668
Train Epoch: 2 [3392/6680 (51%)] Loss: 0.7731
Train Epoch: 2 [3424/6680 (51%)] Loss: 0.7808
Train Epoch: 2 [3456/6680 (52%)] Loss: 0.7867
Train Epoch: 2 [3488/6680 (52%)] Loss: 0.7934
Train Epoch: 2 [3520/6680 (53%)] Loss: 0.7993
Train Epoch: 2 [3552/6680 (53%)] Loss: 0.8051
Train Epoch: 2 [3584/6680 (54%)] Loss: 0.8116
Train Epoch: 2 [3616/6680 (54%)] Loss: 0.8174
Train Epoch: 2 [3648/6680 (55%)] Loss: 0.8232
Train Epoch: 2 [3680/6680 (55%)] Loss: 0.8294
Train Epoch: 2 [3712/6680 (56%)] Loss: 0.8347
Train Epoch: 2 [3744/6680 (56%)] Loss: 0.8405
Train Epoch: 2 [3776/6680 (57%)] Loss: 0.8465
Train Epoch: 2 [3808/6680 (57%)] Loss: 0.8520
Train Epoch: 2 [3840/6680 (57%)] Loss: 0.8583
Train Epoch: 2 [3872/6680 (58%)] Loss: 0.8668
Train Epoch: 2 [3904/6680 (58%)] Loss: 0.8744
Train Epoch: 2 [3936/6680 (59%)] Loss: 0.8806
Train Epoch: 2 [3968/6680 (59%)] Loss: 0.8867
Train Epoch: 2 [4000/6680 (60%)] Loss: 0.8928
Train Epoch: 2 [4032/6680 (60%)] Loss: 0.8977
Train Epoch: 2 [4064/6680 (61%)] Loss: 0.9057
Train Epoch: 2 [4096/6680 (61%)] Loss: 0.9117
Train Epoch: 2 [4128/6680 (62%)] Loss: 0.9165
Train Epoch: 2 [4160/6680 (62%)] Loss: 0.9245
Train Epoch: 2 [4192/6680 (63%)] Loss: 0.9308
Train Epoch: 2 [4224/6680 (63%)] Loss: 0.9365
Train Epoch: 2 [4256/6680 (64%)] Loss: 0.9432
Train Epoch: 2 [4288/6680 (64%)] Loss: 0.9500
Train Epoch: 2 [4320/6680 (65%)] Loss: 0.9562
Train Epoch: 2 [4352/6680 (65%)] Loss: 0.9621
Train Epoch: 2 [4384/6680 (66%)] Loss: 0.9681
Train Epoch: 2 [4416/6680 (66%)] Loss: 0.9737
Train Epoch: 2 [4448/6680 (67%)] Loss: 0.9789
Train Epoch: 2 [4480/6680 (67%)] Loss: 0.9849
Train Epoch: 2 [4512/6680 (68%)] Loss: 0.9904
Train Epoch: 2 [4544/6680 (68%)] Loss: 0.9939
Train Epoch: 2 [4576/6680 (69%)] Loss: 0.9999
Train Epoch: 2 [4608/6680 (69%)] Loss: 1.0059
Train Epoch: 2 [4640/6680 (69%)] Loss: 1.0125
Train Epoch: 2 [4672/6680 (70%)] Loss: 1.0185
Train Epoch: 2 [4704/6680 (70%)] Loss: 1.0242
Train Epoch: 2 [4736/6680 (71%)] Loss: 1.0298
Train Epoch: 2 [4768/6680 (71%)] Loss: 1.0351
Train Epoch: 2 [4800/6680 (72%)] Loss: 1.0420
Train Epoch: 2 [4832/6680 (72%)] Loss: 1.0485
Train Epoch: 2 [4864/6680 (73%)] Loss: 1.0544
Train Epoch: 2 [4896/6680 (73%)] Loss: 1.0619
Train Epoch: 2 [4928/6680 (74%)] Loss: 1.0672
Train Epoch: 2 [4960/6680 (74%)] Loss: 1.0714
Train Epoch: 2 [4992/6680 (75%)] Loss: 1.0768
Train Epoch: 2 [5024/6680 (75%)] Loss: 1.0824
Train Epoch: 2 [5056/6680 (76%)] Loss: 1.0870
Train Epoch: 2 [5088/6680 (76%)] Loss: 1.0936
Train Epoch: 2 [5120/6680 (77%)] Loss: 1.0976
Train Epoch: 2 [5152/6680 (77%)] Loss: 1.1035
Train Epoch: 2 [5184/6680 (78%)] Loss: 1.1090
Train Epoch: 2 [5216/6680 (78%)] Loss: 1.1152
Train Epoch: 2 [5248/6680 (79%)] Loss: 1.1210
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Train Epoch: 2 [5280/6680 (79%)] Loss: 1.1261
Train Epoch: 2 [5312/6680 (80%)] Loss: 1.1302
Train Epoch: 2 [5344/6680 (80%)] Loss: 1.1350
Train Epoch: 2 [5376/6680 (80%)] Loss: 1.1411
Train Epoch: 2 [5408/6680 (81%)] Loss: 1.1466
Train Epoch: 2 [5440/6680 (81%)] Loss: 1.1530
Train Epoch: 2 [5472/6680 (82%)] Loss: 1.1600
Train Epoch: 2 [5504/6680 (82%)] Loss: 1.1649
Train Epoch: 2 [5536/6680 (83%)] Loss: 1.1727
Train Epoch: 2 [5568/6680 (83%)] Loss: 1.1771
Train Epoch: 2 [5600/6680 (84%)] Loss: 1.1828
Train Epoch: 2 [5632/6680 (84%)] Loss: 1.1884
Train Epoch: 2 [5664/6680 (85%)] Loss: 1.1938
Train Epoch: 2 [5696/6680 (85%)] Loss: 1.1998
Train Epoch: 2 [5728/6680 (86%)] Loss: 1.2062
Train Epoch: 2 [5760/6680 (86%)] Loss: 1.2129
Train Epoch: 2 [5792/6680 (87%)] Loss: 1.2180
Train Epoch: 2 [5824/6680 (87%)] Loss: 1.2244
Train Epoch: 2 [5856/6680 (88%)] Loss: 1.2305
Train Epoch: 2 [5888/6680 (88%)] Loss: 1.2358
Train Epoch: 2 [5920/6680 (89%)] Loss: 1.2414
Train Epoch: 2 [5952/6680 (89%)] Loss: 1.2485
Train Epoch: 2 [5984/6680 (90%)] Loss: 1.2536
Train Epoch: 2 [6016/6680 (90%)] Loss: 1.2580
Train Epoch: 2 [6048/6680 (91%)] Loss: 1.2639
Train Epoch: 2 [6080/6680 (91%)] Loss: 1.2698
Train Epoch: 2 [6112/6680 (91%)] Loss: 1.2752
Train Epoch: 2 [6144/6680 (92%)] Loss: 1.2812
Train Epoch: 2 [6176/6680 (92%)] Loss: 1.2872
Train Epoch: 2 [6208/6680 (93%)] Loss: 1.2934
Train Epoch: 2 [6240/6680 (93%)] Loss: 1.2990
Train Epoch: 2 [6272/6680 (94%)] Loss: 1.3044
Train Epoch: 2 [6304/6680 (94%)] Loss: 1.3093
Train Epoch: 2 [6336/6680 (95%)] Loss: 1.3151
Train Epoch: 2 [6368/6680 (95%)] Loss: 1.3206
Train Epoch: 2 [6400/6680 (96%)] Loss: 1.3274
Train Epoch: 2 [6432/6680 (96%)] Loss: 1.3350
Train Epoch: 2 [6464/6680 (97%)] Loss: 1.3411
Train Epoch: 2 [6496/6680 (97%)] Loss: 1.3466
Train Epoch: 2 [6528/6680 (98%)] Loss: 1.3528
Train Epoch: 2 [6560/6680 (98%)] Loss: 1.3596
Train Epoch: 2 [6592/6680 (99%)] Loss: 1.3658
Train Epoch: 2 [6624/6680 (99%)] Loss: 1.3717
Train Epoch: 2 [6656/6680 (100%)] Loss: 1.3774
Train Epoch: 2 [5016/6680 (75%)] Loss: 1.3836
start validating
Validate Epoch: 2 [32/835 (4%)] Loss: 0.0314
Validate Epoch: 2 [64/835 (8%)] Loss: 0.0694
Validate Epoch: 2 [96/835 (11%)] Loss: 0.1137
Validate Epoch: 2 [128/835 (15%)] Loss: 0.1368
Validate Epoch: 2 [160/835 (19%)] Loss: 0.1550
Validate Epoch: 2 [192/835 (23%)] Loss: 0.1900
Validate Epoch: 2 [224/835 (27%)] Loss: 0.2312
Validate Epoch: 2 [256/835 (31%)] Loss: 0.2793
Validate Epoch: 2 [288/835 (34%)] Loss: 0.3161
Validate Epoch: 2 [320/835 (38%)] Loss: 0.3580
Validate Epoch: 2 [352/835 (42%)] Loss: 0.3994
Validate Epoch: 2 [384/835 (46%)] Loss: 0.4353
Validate Epoch: 2 [416/835 (50%)] Loss: 0.4747
Validate Epoch: 2 [448/835 (54%)] Loss: 0.5129
Validate Epoch: 2 [480/835 (57%)] Loss: 0.5616
Validate Epoch: 2 [512/835 (61%)] Loss: 0.6188
Validate Epoch: 2 [544/835 (65%)] Loss: 0.6719
Validate Epoch: 2 [576/835 (69%)] Loss: 0.7215
Validate Epoch: 2 [608/835 (73%)] Loss: 0.7557
Validate Epoch: 2 [640/835 (77%)] Loss: 0.7786
Validate Epoch: 2 [672/835 (80%)] Loss: 0.8286
Validate Epoch: 2 [704/835 (84%)] Loss: 0.8785
Validate Epoch: 2 [736/835 (88%)] Loss: 0.9175
Validate Epoch: 2 [768/835 (92%)] Loss: 0.9613
Validate Epoch: 2 [800/835 (96%)] Loss: 1.0113
Validate Epoch: 2 [832/835 (100%)] Loss: 1.0573
Validate Epoch: 2 [81/835 (10%)] Loss: 1.0883
start training
Train Epoch: 3 [32/6680 (0%)] Loss: 0.0058
Train Epoch: 3 [64/6680 (1%)] Loss: 0.0111
Train Epoch: 3 [96/6680 (1%)] Loss: 0.0170
Train Epoch: 3 [128/6680 (2%)] Loss: 0.0208
Train Epoch: 3 [160/6680 (2%)] Loss: 0.0272
Train Epoch: 3 [192/6680 (3%)] Loss: 0.0326
Train Epoch: 3 [224/6680 (3%)] Loss: 0.0385
Train Epoch: 3 [256/6680 (4%)] Loss: 0.0438
Train Epoch: 3 [288/6680 (4%)] Loss: 0.0476
Train Epoch: 3 [320/6680 (5%)] Loss: 0.0522
```

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Train Epoch: 3 [352/6680 (5%)] Loss: 0.0567
Train Epoch: 3 [384/6680 (6%)] Loss: 0.0620
Train Epoch: 3 [416/6680 (6%)] Loss: 0.0686
Train Epoch: 3 [448/6680 (7%)] Loss: 0.0733
Train Epoch: 3 [480/6680 (7%)] Loss: 0.0791
Train Epoch: 3 [512/6680 (8%)] Loss: 0.0858
Train Epoch: 3 [544/6680 (8%)] Loss: 0.0899
Train Epoch: 3 [576/6680 (9%)] Loss: 0.0955
Train Epoch: 3 [608/6680 (9%)] Loss: 0.1010
Train Epoch: 3 [640/6680 (10%)] Loss: 0.1059
Train Epoch: 3 [672/6680 (10%)] Loss: 0.1113
Train Epoch: 3 [704/6680 (11%)] Loss: 0.1151
Train Epoch: 3 [736/6680 (11%)] Loss: 0.1209
Train Epoch: 3 [768/6680 (11%)] Loss: 0.1255
Train Epoch: 3 [800/6680 (12%)] Loss: 0.1302
Train Epoch: 3 [832/6680 (12%)] Loss: 0.1353
Train Epoch: 3 [864/6680 (13%)] Loss: 0.1412
Train Epoch: 3 [896/6680 (13%)] Loss: 0.1460
Train Epoch: 3 [928/6680 (14%)] Loss: 0.1520
Train Epoch: 3 [960/6680 (14%)] Loss: 0.1561
Train Epoch: 3 [992/6680 (15%)] Loss: 0.1606
Train Epoch: 3 [1024/6680 (15%)] Loss: 0.1672
Train Epoch: 3 [1056/6680 (16%)] Loss: 0.1732
Train Epoch: 3 [1088/6680 (16%)] Loss: 0.1792
Train Epoch: 3 [1120/6680 (17%)] Loss: 0.1849
Train Epoch: 3 [1152/6680 (17%)] Loss: 0.1895
Train Epoch: 3 [1184/6680 (18%)] Loss: 0.1938
Train Epoch: 3 [1216/6680 (18%)] Loss: 0.2005
Train Epoch: 3 [1248/6680 (19%)] Loss: 0.2050
Train Epoch: 3 [1280/6680 (19%)] Loss: 0.2094
Train Epoch: 3 [1312/6680 (20%)] Loss: 0.2150
Train Epoch: 3 [1344/6680 (20%)] Loss: 0.2201
Train Epoch: 3 [1376/6680 (21%)] Loss: 0.2259
Train Epoch: 3 [1408/6680 (21%)] Loss: 0.2302
Train Epoch: 3 [1440/6680 (22%)] Loss: 0.2361
Train Epoch: 3 [1472/6680 (22%)] Loss: 0.2406
Train Epoch: 3 [1504/6680 (23%)] Loss: 0.2456
Train Epoch: 3 [1536/6680 (23%)] Loss: 0.2492
Train Epoch: 3 [1568/6680 (23%)] Loss: 0.2542
Train Epoch: 3 [1600/6680 (24%)] Loss: 0.2585
Train Epoch: 3 [1632/6680 (24%)] Loss: 0.2650
Train Epoch: 3 [1664/6680 (25%)] Loss: 0.2708
Train Epoch: 3 [1696/6680 (25%)] Loss: 0.2748
Train Epoch: 3 [1728/6680 (26%)] Loss: 0.2797
Train Epoch: 3 [1760/6680 (26%)] Loss: 0.2848
Train Epoch: 3 [1792/6680 (27%)] Loss: 0.2897
Train Epoch: 3 [1824/6680 (27%)] Loss: 0.2943
Train Epoch: 3 [1856/6680 (28%)] Loss: 0.2993
Train Epoch: 3 [1888/6680 (28%)] Loss: 0.3031
Train Epoch: 3 [1920/6680 (29%)] Loss: 0.3078
Train Epoch: 3 [1952/6680 (29%)] Loss: 0.3119
Train Epoch: 3 [1984/6680 (30%)] Loss: 0.3171
Train Epoch: 3 [2016/6680 (30%)] Loss: 0.3224
Train Epoch: 3 [2048/6680 (31%)] Loss: 0.3274
Train Epoch: 3 [2080/6680 (31%)] Loss: 0.3323
Train Epoch: 3 [2112/6680 (32%)] Loss: 0.3356
Train Epoch: 3 [2144/6680 (32%)] Loss: 0.3416
Train Epoch: 3 [2176/6680 (33%)] Loss: 0.3452
Train Epoch: 3 [2208/6680 (33%)] Loss: 0.3492
Train Epoch: 3 [2240/6680 (34%)] Loss: 0.3551
Train Epoch: 3 [2272/6680 (34%)] Loss: 0.3607
Train Epoch: 3 [2304/6680 (34%)] Loss: 0.3644
Train Epoch: 3 [2336/6680 (35%)] Loss: 0.3694
Train Epoch: 3 [2368/6680 (35%)] Loss: 0.3735
Train Epoch: 3 [2400/6680 (36%)] Loss: 0.3785
Train Epoch: 3 [2432/6680 (36%)] Loss: 0.3844
Train Epoch: 3 [2464/6680 (37%)] Loss: 0.3891
Train Epoch: 3 [2496/6680 (37%)] Loss: 0.3975
Train Epoch: 3 [2528/6680 (38%)] Loss: 0.4026
Train Epoch: 3 [2560/6680 (38%)] Loss: 0.4078
Train Epoch: 3 [2592/6680 (39%)] Loss: 0.4121
Train Epoch: 3 [2624/6680 (39%)] Loss: 0.4158
Train Epoch: 3 [2656/6680 (40%)] Loss: 0.4214
Train Epoch: 3 [2688/6680 (40%)] Loss: 0.4273
Train Epoch: 3 [2720/6680 (41%)] Loss: 0.4329
Train Epoch: 3 [2752/6680 (41%)] Loss: 0.4385
Train Epoch: 3 [2784/6680 (42%)] Loss: 0.4455
Train Epoch: 3 [2816/6680 (42%)] Loss: 0.4508
Train Epoch: 3 [2848/6680 (43%)] Loss: 0.4566
Train Epoch: 3 [2880/6680 (43%)] Loss: 0.4620
Train Epoch: 3 [2912/6680 (44%)] Loss: 0.4667
Train Epoch: 3 [2944/6680 (44%)] Loss: 0.4717
Train Epoch: 3 [2976/6680 (45%)] Loss: 0.4775
Train Epoch: 3 [3008/6680 (45%)] Loss: 0.4825
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Train Epoch: 3 [3040/6680 (46%)] Loss: 0.4866
Train Epoch: 3 [3072/6680 (46%)] Loss: 0.4920
Train Epoch: 3 [3104/6680 (46%)] Loss: 0.4968
Train Epoch: 3 [3136/6680 (47%)] Loss: 0.5002
Train Epoch: 3 [3168/6680 (47%)] Loss: 0.5067
Train Epoch: 3 [3200/6680 (48%)] Loss: 0.5131
Train Epoch: 3 [3232/6680 (48%)] Loss: 0.5173
Train Epoch: 3 [3264/6680 (49%)] Loss: 0.5232
Train Epoch: 3 [3296/6680 (49%)] Loss: 0.5270
Train Epoch: 3 [3328/6680 (50%)] Loss: 0.5319
Train Epoch: 3 [3360/6680 (50%)] Loss: 0.5363
Train Epoch: 3 [3392/6680 (51%)] Loss: 0.5402
Train Epoch: 3 [3424/6680 (51%)] Loss: 0.5457
Train Epoch: 3 [3456/6680 (52%)] Loss: 0.5515
Train Epoch: 3 [3488/6680 (52%)] Loss: 0.5566
Train Epoch: 3 [3520/6680 (53%)] Loss: 0.5613
Train Epoch: 3 [3552/6680 (53%)] Loss: 0.5663
Train Epoch: 3 [3584/6680 (54%)] Loss: 0.5700
Train Epoch: 3 [3616/6680 (54%)] Loss: 0.5748
Train Epoch: 3 [3648/6680 (55%)] Loss: 0.5821
Train Epoch: 3 [3680/6680 (55%)] Loss: 0.5871
Train Epoch: 3 [3712/6680 (56%)] Loss: 0.5903
Train Epoch: 3 [3744/6680 (56%)] Loss: 0.5959
Train Epoch: 3 [3776/6680 (57%)] Loss: 0.5999
Train Epoch: 3 [3808/6680 (57%)] Loss: 0.6039
Train Epoch: 3 [3840/6680 (57%)] Loss: 0.6093
Train Epoch: 3 [3872/6680 (58%)] Loss: 0.6143
Train Epoch: 3 [3904/6680 (58%)] Loss: 0.6193
Train Epoch: 3 [3936/6680 (59%)] Loss: 0.6244
Train Epoch: 3 [3968/6680 (59%)] Loss: 0.6280
Train Epoch: 3 [4000/6680 (60%)] Loss: 0.6313
Train Epoch: 3 [4032/6680 (60%)] Loss: 0.6370
Train Epoch: 3 [4064/6680 (61%)] Loss: 0.6412
Train Epoch: 3 [4096/6680 (61%)] Loss: 0.6472
Train Epoch: 3 [4128/6680 (62%)] Loss: 0.6497
Train Epoch: 3 [4160/6680 (62%)] Loss: 0.6539
Train Epoch: 3 [4192/6680 (63%)] Loss: 0.6588
Train Epoch: 3 [4224/6680 (63%)] Loss: 0.6630
Train Epoch: 3 [4256/6680 (64%)] Loss: 0.6684
Train Epoch: 3 [4288/6680 (64%)] Loss: 0.6729
Train Epoch: 3 [4320/6680 (65%)] Loss: 0.6785
Train Epoch: 3 [4352/6680 (65%)] Loss: 0.6819
Train Epoch: 3 [4384/6680 (66%)] Loss: 0.6861
Train Epoch: 3 [4416/6680 (66%)] Loss: 0.6920
Train Epoch: 3 [4448/6680 (67%)] Loss: 0.6969
Train Epoch: 3 [4480/6680 (67%)] Loss: 0.7023
Train Epoch: 3 [4512/6680 (68%)] Loss: 0.7068
Train Epoch: 3 [4544/6680 (68%)] Loss: 0.7118
Train Epoch: 3 [4576/6680 (69%)] Loss: 0.7160
Train Epoch: 3 [4608/6680 (69%)] Loss: 0.7193
Train Epoch: 3 [4640/6680 (69%)] Loss: 0.7250
Train Epoch: 3 [4672/6680 (70%)] Loss: 0.7295
Train Epoch: 3 [4704/6680 (70%)] Loss: 0.7334
Train Epoch: 3 [4736/6680 (71%)] Loss: 0.7385
Train Epoch: 3 [4768/6680 (71%)] Loss: 0.7424
Train Epoch: 3 [4800/6680 (72%)] Loss: 0.7460
Train Epoch: 3 [4832/6680 (72%)] Loss: 0.7508
Train Epoch: 3 [4864/6680 (73%)] Loss: 0.7540
Train Epoch: 3 [4896/6680 (73%)] Loss: 0.7596
Train Epoch: 3 [4928/6680 (74%)] Loss: 0.7663
Train Epoch: 3 [4960/6680 (74%)] Loss: 0.7706
Train Epoch: 3 [4992/6680 (75%)] Loss: 0.7745
Train Epoch: 3 [5024/6680 (75%)] Loss: 0.7792
Train Epoch: 3 [5056/6680 (76%)] Loss: 0.7833
Train Epoch: 3 [5088/6680 (76%)] Loss: 0.7874
Train Epoch: 3 [5120/6680 (77%)] Loss: 0.7922
Train Epoch: 3 [5152/6680 (77%)] Loss: 0.7976
Train Epoch: 3 [5184/6680 (78%)] Loss: 0.8015
Train Epoch: 3 [5216/6680 (78%)] Loss: 0.8052
Train Epoch: 3 [5248/6680 (79%)] Loss: 0.8091
Train Epoch: 3 [5280/6680 (79%)] Loss: 0.8148
Train Epoch: 3 [5312/6680 (80%)] Loss: 0.8193
Train Epoch: 3 [5344/6680 (80%)] Loss: 0.8237
Train Epoch: 3 [5376/6680 (80%)] Loss: 0.8296
Train Epoch: 3 [5408/6680 (81%)] Loss: 0.8338
Train Epoch: 3 [5440/6680 (81%)] Loss: 0.8383
Train Epoch: 3 [5472/6680 (82%)] Loss: 0.8450
Train Epoch: 3 [5504/6680 (82%)] Loss: 0.8490
Train Epoch: 3 [5536/6680 (83%)] Loss: 0.8543
Train Epoch: 3 [5568/6680 (83%)] Loss: 0.8603
Train Epoch: 3 [5600/6680 (84%)] Loss: 0.8640
Train Epoch: 3 [5632/6680 (84%)] Loss: 0.8677
Train Epoch: 3 [5664/6680 (85%)] Loss: 0.8744
Train Epoch: 3 [5696/6680 (85%)] Loss: 0.8784
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Train Epoch: 3 [5728/6680 (86%)] Loss: 0.8839
Train Epoch: 3 [5760/6680 (86%)] Loss: 0.8889
Train Epoch: 3 [5792/6680 (87%)] Loss: 0.8931
Train Epoch: 3 [5824/6680 (87%)] Loss: 0.8973
Train Epoch: 3 [5856/6680 (88%)] Loss: 0.9032
Train Epoch: 3 [5888/6680 (88%)] Loss: 0.9078
Train Epoch: 3 [5920/6680 (89%)] Loss: 0.9119
Train Epoch: 3 [5952/6680 (89%)] Loss: 0.9178
Train Epoch: 3 [5984/6680 (90%)] Loss: 0.9235
Train Epoch: 3 [6016/6680 (90%)] Loss: 0.9287
Train Epoch: 3 [6048/6680 (91%)] Loss: 0.9324
Train Epoch: 3 [6080/6680 (91%)] Loss: 0.9363
Train Epoch: 3 [6112/6680 (91%)] Loss: 0.9416
Train Epoch: 3 [6144/6680 (92%)] Loss: 0.9463
Train Epoch: 3 [6176/6680 (92%)] Loss: 0.9499
Train Epoch: 3 [6208/6680 (93%)] Loss: 0.9560
Train Epoch: 3 [6240/6680 (93%)] Loss: 0.9605
Train Epoch: 3 [6272/6680 (94%)] Loss: 0.9644
Train Epoch: 3 [6304/6680 (94%)] Loss: 0.9690
Train Epoch: 3 [6336/6680 (95%)] Loss: 0.9728
Train Epoch: 3 [6368/6680 (95%)] Loss: 0.9780
Train Epoch: 3 [6400/6680 (96%)] Loss: 0.9821
Train Epoch: 3 [6432/6680 (96%)] Loss: 0.9868
Train Epoch: 3 [6464/6680 (97%)] Loss: 0.9917
Train Epoch: 3 [6496/6680 (97%)] Loss: 0.9945
Train Epoch: 3 [6528/6680 (98%)] Loss: 0.9988
Train Epoch: 3 [6560/6680 (98%)] Loss: 1.0052
Train Epoch: 3 [6592/6680 (99%)] Loss: 1.0107
Train Epoch: 3 [6624/6680 (99%)] Loss: 1.0157
Train Epoch: 3 [6656/6680 (100%)] Loss: 1.0216
Train Epoch: 3 [5016/6680 (75%)] Loss: 1.0256
start validating
Validate Epoch: 3 [32/835 (4%)] Loss: 0.0215
Validate Epoch: 3 [64/835 (8%)] Loss: 0.0516
Validate Epoch: 3 [96/835 (11%)] Loss: 0.0863
Validate Epoch: 3 [128/835 (15%)] Loss: 0.1090
Validate Epoch: 3 [160/835 (19%)] Loss: 0.1285
Validate Epoch: 3 [192/835 (23%)] Loss: 0.1541
Validate Epoch: 3 [224/835 (27%)] Loss: 0.1928
Validate Epoch: 3 [256/835 (31%)] Loss: 0.2351
Validate Epoch: 3 [288/835 (34%)] Loss: 0.2729
Validate Epoch: 3 [320/835 (38%)] Loss: 0.3042
Validate Epoch: 3 [352/835 (42%)] Loss: 0.3423
Validate Epoch: 3 [384/835 (46%)] Loss: 0.3756
Validate Epoch: 3 [416/835 (50%)] Loss: 0.4185
Validate Epoch: 3 [448/835 (54%)] Loss: 0.4478
Validate Epoch: 3 [480/835 (57%)] Loss: 0.4898
Validate Epoch: 3 [512/835 (61%)] Loss: 0.5372
Validate Epoch: 3 [544/835 (65%)] Loss: 0.5826
Validate Epoch: 3 [576/835 (69%)] Loss: 0.6282
Validate Epoch: 3 [608/835 (73%)] Loss: 0.6537
Validate Epoch: 3 [640/835 (77%)] Loss: 0.6807
Validate Epoch: 3 [672/835 (80%)] Loss: 0.7214
Validate Epoch: 3 [704/835 (84%)] Loss: 0.7584
Validate Epoch: 3 [736/835 (88%)] Loss: 0.7947
Validate Epoch: 3 [768/835 (92%)] Loss: 0.8330
Validate Epoch: 3 [800/835 (96%)] Loss: 0.8722
Validate Epoch: 3 [832/835 (100%)] Loss: 0.9043
Validate Epoch: 3 [81/835 (10%)] Loss: 0.9533
start training
Train Epoch: 4 [32/6680 (0%)] Loss: 0.0040
Train Epoch: 4 [64/6680 (1%)] Loss: 0.0089
Train Epoch: 4 [96/6680 (1%)] Loss: 0.0139
Train Epoch: 4 [128/6680 (2%)] Loss: 0.0188
Train Epoch: 4 [160/6680 (2%)] Loss: 0.0243
Train Epoch: 4 [192/6680 (3%)] Loss: 0.0276
Train Epoch: 4 [224/6680 (3%)] Loss: 0.0316
Train Epoch: 4 [256/6680 (4%)] Loss: 0.0359
Train Epoch: 4 [288/6680 (4%)] Loss: 0.0399
Train Epoch: 4 [320/6680 (5%)] Loss: 0.0429
Train Epoch: 4 [352/6680 (5%)] Loss: 0.0473
Train Epoch: 4 [384/6680 (6%)] Loss: 0.0526
Train Epoch: 4 [416/6680 (6%)] Loss: 0.0566
Train Epoch: 4 [448/6680 (7%)] Loss: 0.0611
Train Epoch: 4 [480/6680 (7%)] Loss: 0.0650
Train Epoch: 4 [512/6680 (8%)] Loss: 0.0700
Train Epoch: 4 [544/6680 (8%)] Loss: 0.0729
Train Epoch: 4 [576/6680 (9%)] Loss: 0.0768
Train Epoch: 4 [608/6680 (9%)] Loss: 0.0807
Train Epoch: 4 [640/6680 (10%)] Loss: 0.0856
Train Epoch: 4 [672/6680 (10%)] Loss: 0.0909
Train Epoch: 4 [704/6680 (11%)] Loss: 0.0943
Train Epoch: 4 [736/6680 (11%)] Loss: 0.0998
Train Epoch: 4 [768/6680 (11%)] Loss: 0.1033
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Train Epoch: 4 [800/6680 (12%)] Loss: 0.1086
Train Epoch: 4 [832/6680 (12%)] Loss: 0.1135
Train Epoch: 4 [864/6680 (13%)] Loss: 0.1179
Train Epoch: 4 [896/6680 (13%)] Loss: 0.1216
Train Epoch: 4 [928/6680 (14%)] Loss: 0.1266
Train Epoch: 4 [960/6680 (14%)] Loss: 0.1326
Train Epoch: 4 [992/6680 (15%)] Loss: 0.1368
Train Epoch: 4 [1024/6680 (15%)] Loss: 0.1419
Train Epoch: 4 [1056/6680 (16%)] Loss: 0.1457
Train Epoch: 4 [1088/6680 (16%)] Loss: 0.1490
Train Epoch: 4 [1120/6680 (17%)] Loss: 0.1528
Train Epoch: 4 [1152/6680 (17%)] Loss: 0.1576
Train Epoch: 4 [1184/6680 (18%)] Loss: 0.1617
Train Epoch: 4 [1216/6680 (18%)] Loss: 0.1668
Train Epoch: 4 [1248/6680 (19%)] Loss: 0.1712
Train Epoch: 4 [1280/6680 (19%)] Loss: 0.1749
Train Epoch: 4 [1312/6680 (20%)] Loss: 0.1782
Train Epoch: 4 [1344/6680 (20%)] Loss: 0.1830
Train Epoch: 4 [1376/6680 (21%)] Loss: 0.1882
Train Epoch: 4 [1408/6680 (21%)] Loss: 0.1933
Train Epoch: 4 [1440/6680 (22%)] Loss: 0.1964
Train Epoch: 4 [1472/6680 (22%)] Loss: 0.2019
Train Epoch: 4 [1504/6680 (23%)] Loss: 0.2071
Train Epoch: 4 [1536/6680 (23%)] Loss: 0.2123
Train Epoch: 4 [1568/6680 (23%)] Loss: 0.2174
Train Epoch: 4 [1600/6680 (24%)] Loss: 0.2219
Train Epoch: 4 [1632/6680 (24%)] Loss: 0.2267
Train Epoch: 4 [1664/6680 (25%)] Loss: 0.2307
Train Epoch: 4 [1696/6680 (25%)] Loss: 0.2350
Train Epoch: 4 [1728/6680 (26%)] Loss: 0.2410
Train Epoch: 4 [1760/6680 (26%)] Loss: 0.2470
Train Epoch: 4 [1792/6680 (27%)] Loss: 0.2512
Train Epoch: 4 [1824/6680 (27%)] Loss: 0.2552
Train Epoch: 4 [1856/6680 (28%)] Loss: 0.2590
Train Epoch: 4 [1888/6680 (28%)] Loss: 0.2633
Train Epoch: 4 [1920/6680 (29%)] Loss: 0.2675
Train Epoch: 4 [1952/6680 (29%)] Loss: 0.2711
Train Epoch: 4 [1984/6680 (30%)] Loss: 0.2760
Train Epoch: 4 [2016/6680 (30%)] Loss: 0.2793
Train Epoch: 4 [2048/6680 (31%)] Loss: 0.2825
Train Epoch: 4 [2080/6680 (31%)] Loss: 0.2873
Train Epoch: 4 [2112/6680 (32%)] Loss: 0.2904
Train Epoch: 4 [2144/6680 (32%)] Loss: 0.2935
Train Epoch: 4 [2176/6680 (33%)] Loss: 0.2974
Train Epoch: 4 [2208/6680 (33%)] Loss: 0.3026
Train Epoch: 4 [2240/6680 (34%)] Loss: 0.3067
Train Epoch: 4 [2272/6680 (34%)] Loss: 0.3123
Train Epoch: 4 [2304/6680 (34%)] Loss: 0.3169
Train Epoch: 4 [2336/6680 (35%)] Loss: 0.3218
Train Epoch: 4 [2368/6680 (35%)] Loss: 0.3265
Train Epoch: 4 [2400/6680 (36%)] Loss: 0.3305
Train Epoch: 4 [2432/6680 (36%)] Loss: 0.3346
Train Epoch: 4 [2464/6680 (37%)] Loss: 0.3400
Train Epoch: 4 [2496/6680 (37%)] Loss: 0.3437
Train Epoch: 4 [2528/6680 (38%)] Loss: 0.3480
Train Epoch: 4 [2560/6680 (38%)] Loss: 0.3520
Train Epoch: 4 [2592/6680 (39%)] Loss: 0.3558
Train Epoch: 4 [2624/6680 (39%)] Loss: 0.3596
Train Epoch: 4 [2656/6680 (40%)] Loss: 0.3629
Train Epoch: 4 [2688/6680 (40%)] Loss: 0.3672
Train Epoch: 4 [2720/6680 (41%)] Loss: 0.3706
Train Epoch: 4 [2752/6680 (41%)] Loss: 0.3749
Train Epoch: 4 [2784/6680 (42%)] Loss: 0.3787
Train Epoch: 4 [2816/6680 (42%)] Loss: 0.3831
Train Epoch: 4 [2848/6680 (43%)] Loss: 0.3888
Train Epoch: 4 [2880/6680 (43%)] Loss: 0.3915
Train Epoch: 4 [2912/6680 (44%)] Loss: 0.3949
Train Epoch: 4 [2944/6680 (44%)] Loss: 0.3981
Train Epoch: 4 [2976/6680 (45%)] Loss: 0.4027
Train Epoch: 4 [3008/6680 (45%)] Loss: 0.4074
Train Epoch: 4 [3040/6680 (46%)] Loss: 0.4114
Train Epoch: 4 [3072/6680 (46%)] Loss: 0.4158
Train Epoch: 4 [3104/6680 (46%)] Loss: 0.4204
Train Epoch: 4 [3136/6680 (47%)] Loss: 0.4229
Train Epoch: 4 [3168/6680 (47%)] Loss: 0.4268
Train Epoch: 4 [3200/6680 (48%)] Loss: 0.4310
Train Epoch: 4 [3232/6680 (48%)] Loss: 0.4348
Train Epoch: 4 [3264/6680 (49%)] Loss: 0.4397
Train Epoch: 4 [3296/6680 (49%)] Loss: 0.4442
Train Epoch: 4 [3328/6680 (50%)] Loss: 0.4483
Train Epoch: 4 [3360/6680 (50%)] Loss: 0.4519
Train Epoch: 4 [3392/6680 (51%)] Loss: 0.4558
Train Epoch: 4 [3424/6680 (51%)] Loss: 0.4595
Train Epoch: 4 [3456/6680 (52%)] Loss: 0.4632

Train Epoch: 4 [3488/6680 (52%)] Loss: 0.4675
Train Epoch: 4 [3520/6680 (53%)] Loss: 0.4711
Train Epoch: 4 [3552/6680 (53%)] Loss: 0.4756
Train Epoch: 4 [3584/6680 (54%)] Loss: 0.4795
Train Epoch: 4 [3616/6680 (54%)] Loss: 0.4856
Train Epoch: 4 [3648/6680 (55%)] Loss: 0.4891
Train Epoch: 4 [3680/6680 (55%)] Loss: 0.4962
Train Epoch: 4 [3712/6680 (56%)] Loss: 0.5008
Train Epoch: 4 [3744/6680 (56%)] Loss: 0.5067
Train Epoch: 4 [3776/6680 (57%)] Loss: 0.5105
Train Epoch: 4 [3808/6680 (57%)] Loss: 0.5144
Train Epoch: 4 [3840/6680 (57%)] Loss: 0.5193
Train Epoch: 4 [3872/6680 (58%)] Loss: 0.5232
Train Epoch: 4 [3904/6680 (58%)] Loss: 0.5275
Train Epoch: 4 [3936/6680 (59%)] Loss: 0.5315
Train Epoch: 4 [3968/6680 (59%)] Loss: 0.5355
Train Epoch: 4 [4000/6680 (60%)] Loss: 0.5403
Train Epoch: 4 [4032/6680 (60%)] Loss: 0.5438
Train Epoch: 4 [4064/6680 (61%)] Loss: 0.5484
Train Epoch: 4 [4096/6680 (61%)] Loss: 0.5528

VanishingGradient: InProgress

Overfit: InProgress

Overtraining: IssuesFound

PoorWeightInitialization: IssuesFound

LossNotDecreasing: InProgress

Train Epoch: 4 [4128/6680 (62%)] Loss: 0.5577
Train Epoch: 4 [4160/6680 (62%)] Loss: 0.5625
Train Epoch: 4 [4192/6680 (63%)] Loss: 0.5664
Train Epoch: 4 [4224/6680 (63%)] Loss: 0.5708
Train Epoch: 4 [4256/6680 (64%)] Loss: 0.5741
Train Epoch: 4 [4288/6680 (64%)] Loss: 0.5789
Train Epoch: 4 [4320/6680 (65%)] Loss: 0.5826
Train Epoch: 4 [4352/6680 (65%)] Loss: 0.5857
Train Epoch: 4 [4384/6680 (66%)] Loss: 0.5901
Train Epoch: 4 [4416/6680 (66%)] Loss: 0.5949
Train Epoch: 4 [4448/6680 (67%)] Loss: 0.6011
Train Epoch: 4 [4480/6680 (67%)] Loss: 0.6037
Train Epoch: 4 [4512/6680 (68%)] Loss: 0.6074
Train Epoch: 4 [4544/6680 (68%)] Loss: 0.6105
Train Epoch: 4 [4576/6680 (69%)] Loss: 0.6144
Train Epoch: 4 [4608/6680 (69%)] Loss: 0.6184
Train Epoch: 4 [4640/6680 (69%)] Loss: 0.6227
Train Epoch: 4 [4672/6680 (70%)] Loss: 0.6274
Train Epoch: 4 [4704/6680 (70%)] Loss: 0.6313
Train Epoch: 4 [4736/6680 (71%)] Loss: 0.6358
Train Epoch: 4 [4768/6680 (71%)] Loss: 0.6408
Train Epoch: 4 [4800/6680 (72%)] Loss: 0.6447
Train Epoch: 4 [4832/6680 (72%)] Loss: 0.6476
Train Epoch: 4 [4864/6680 (73%)] Loss: 0.6517
Train Epoch: 4 [4896/6680 (73%)] Loss: 0.6561
Train Epoch: 4 [4928/6680 (74%)] Loss: 0.6606
Train Epoch: 4 [4960/6680 (74%)] Loss: 0.6669
Train Epoch: 4 [4992/6680 (75%)] Loss: 0.6706
Train Epoch: 4 [5024/6680 (75%)] Loss: 0.6745
Train Epoch: 4 [5056/6680 (76%)] Loss: 0.6794
Train Epoch: 4 [5088/6680 (76%)] Loss: 0.6855
Train Epoch: 4 [5120/6680 (77%)] Loss: 0.6892
Train Epoch: 4 [5152/6680 (77%)] Loss: 0.6927
Train Epoch: 4 [5184/6680 (78%)] Loss: 0.6964
Train Epoch: 4 [5216/6680 (78%)] Loss: 0.7008
Train Epoch: 4 [5248/6680 (79%)] Loss: 0.7041
Train Epoch: 4 [5280/6680 (79%)] Loss: 0.7067
Train Epoch: 4 [5312/6680 (80%)] Loss: 0.7113
Train Epoch: 4 [5344/6680 (80%)] Loss: 0.7163
Train Epoch: 4 [5376/6680 (80%)] Loss: 0.7214
Train Epoch: 4 [5408/6680 (81%)] Loss: 0.7250
Train Epoch: 4 [5440/6680 (81%)] Loss: 0.7295
Train Epoch: 4 [5472/6680 (82%)] Loss: 0.7328
Train Epoch: 4 [5504/6680 (82%)] Loss: 0.7365
Train Epoch: 4 [5536/6680 (83%)] Loss: 0.7406
Train Epoch: 4 [5568/6680 (83%)] Loss: 0.7450
Train Epoch: 4 [5600/6680 (84%)] Loss: 0.7511
Train Epoch: 4 [5632/6680 (84%)] Loss: 0.7561
Train Epoch: 4 [5664/6680 (85%)] Loss: 0.7622
Train Epoch: 4 [5696/6680 (85%)] Loss: 0.7657
Train Epoch: 4 [5728/6680 (86%)] Loss: 0.7692
Train Epoch: 4 [5760/6680 (86%)] Loss: 0.7735
Train Epoch: 4 [5792/6680 (87%)] Loss: 0.7781
Train Epoch: 4 [5824/6680 (87%)] Loss: 0.7804
Train Epoch: 4 [5856/6680 (88%)] Loss: 0.7833
Train Epoch: 4 [5888/6680 (88%)] Loss: 0.7865
Train Epoch: 4 [5920/6680 (89%)] Loss: 0.7896
Train Epoch: 4 [5952/6680 (89%)] Loss: 0.7953
Train Epoch: 4 [5984/6680 (90%)] Loss: 0.7984

```
Train Epoch: 4 [6016/6680 (90%)] Loss: 0.8016
Train Epoch: 4 [6048/6680 (91%)] Loss: 0.8057
Train Epoch: 4 [6080/6680 (91%)] Loss: 0.8092
Train Epoch: 4 [6112/6680 (91%)] Loss: 0.8141
Train Epoch: 4 [6144/6680 (92%)] Loss: 0.8175
Train Epoch: 4 [6176/6680 (92%)] Loss: 0.8212
Train Epoch: 4 [6208/6680 (93%)] Loss: 0.8264
Train Epoch: 4 [6240/6680 (93%)] Loss: 0.8301
Train Epoch: 4 [6272/6680 (94%)] Loss: 0.8326
Train Epoch: 4 [6304/6680 (94%)] Loss: 0.8357
Train Epoch: 4 [6336/6680 (95%)] Loss: 0.8408
Train Epoch: 4 [6368/6680 (95%)] Loss: 0.8437
Train Epoch: 4 [6400/6680 (96%)] Loss: 0.8469
Train Epoch: 4 [6432/6680 (96%)] Loss: 0.8502
Train Epoch: 4 [6464/6680 (97%)] Loss: 0.8536
Train Epoch: 4 [6496/6680 (97%)] Loss: 0.8585
Train Epoch: 4 [6528/6680 (98%)] Loss: 0.8638
Train Epoch: 4 [6560/6680 (98%)] Loss: 0.8680
Train Epoch: 4 [6592/6680 (99%)] Loss: 0.8714
Train Epoch: 4 [6624/6680 (99%)] Loss: 0.8744
Train Epoch: 4 [6656/6680 (100%)] Loss: 0.8786
Train Epoch: 4 [5016/6680 (75%)] Loss: 0.8839
start validating
Validate Epoch: 4 [32/835 (4%)] Loss: 0.0148
Validate Epoch: 4 [64/835 (8%)] Loss: 0.0421
Validate Epoch: 4 [96/835 (11%)] Loss: 0.0722
Validate Epoch: 4 [128/835 (15%)] Loss: 0.0866
Validate Epoch: 4 [160/835 (19%)] Loss: 0.0980
Validate Epoch: 4 [192/835 (23%)] Loss: 0.1247
Validate Epoch: 4 [224/835 (27%)] Loss: 0.1632
Validate Epoch: 4 [256/835 (31%)] Loss: 0.1942
Validate Epoch: 4 [288/835 (34%)] Loss: 0.2287
Validate Epoch: 4 [320/835 (38%)] Loss: 0.2686
Validate Epoch: 4 [352/835 (42%)] Loss: 0.3064
Validate Epoch: 4 [384/835 (46%)] Loss: 0.3339
Validate Epoch: 4 [416/835 (50%)] Loss: 0.3701
Validate Epoch: 4 [448/835 (54%)] Loss: 0.3936
Validate Epoch: 4 [480/835 (57%)] Loss: 0.4311
Validate Epoch: 4 [512/835 (61%)] Loss: 0.4709
Validate Epoch: 4 [544/835 (65%)] Loss: 0.5182
Validate Epoch: 4 [576/835 (69%)] Loss: 0.5622
Validate Epoch: 4 [608/835 (73%)] Loss: 0.5824
Validate Epoch: 4 [640/835 (77%)] Loss: 0.6062
Validate Epoch: 4 [672/835 (80%)] Loss: 0.6502

2024-08-17 09:08:22 Uploading - Uploading generated training modelValidate Epoch: 4 [704/835 (84%)] Loss: 0.6910
Validate Epoch: 4 [736/835 (88%)] Loss: 0.7144
Validate Epoch: 4 [768/835 (92%)] Loss: 0.7464
Validate Epoch: 4 [800/835 (96%)] Loss: 0.7907
Validate Epoch: 4 [832/835 (100%)] Loss: 0.8274
Validate Epoch: 4 [81/835 (10%)] Loss: 0.8826
Saving the model.
Downloading: "https://download.pytorch.org/models/resnet18-5c106cde.pth" to /root/.cache/torch/hub/checkpoints/resnet18-5c106cde.pth
#015 0%| 0.00/44.7M [00:00<?, ?B/s]#015 18%| 8.16M/44.7M [00:00<00:00, 85.5MB/s]#015 37%| 16.5M/44.7M [00:00<00:00, 86.5MB/s]#015 56%| 24.9M/44.7M [00:00<00:00, 87.0MB/s]#015 74%| 33.2M/44.7M [00:00<00:00, 87.2MB/s]#015 94%| 41.8M/44.7M [00:00<00:00, 88.3MB/s]#015 100%| 44.7M/44.7M [00:00<00:00, 87.7MB/s]
INFO:__main__:Train Epoch: 1 [32/6680 (0%)] Loss: 0.0233
INFO:__main__:Train Epoch: 1 [64/6680 (1%)] Loss: 0.0729
INFO:__main__:Train Epoch: 1 [96/6680 (1%)] Loss: 0.1024
INFO:__main__:Train Epoch: 1 [128/6680 (2%)] Loss: 0.1314
INFO:__main__:Train Epoch: 1 [160/6680 (2%)] Loss: 0.1548
INFO:__main__:Train Epoch: 1 [192/6680 (3%)] Loss: 0.1790
INFO:__main__:Train Epoch: 1 [224/6680 (3%)] Loss: 0.2027
INFO:__main__:Train Epoch: 1 [256/6680 (4%)] Loss: 0.2259
INFO:__main__:Train Epoch: 1 [288/6680 (4%)] Loss: 0.2486
INFO:__main__:Train Epoch: 1 [320/6680 (5%)] Loss: 0.2716
INFO:__main__:Train Epoch: 1 [352/6680 (5%)] Loss: 0.2946
INFO:__main__:Train Epoch: 1 [384/6680 (6%)] Loss: 0.3178
INFO:__main__:Train Epoch: 1 [416/6680 (6%)] Loss: 0.3411
INFO:__main__:Train Epoch: 1 [448/6680 (7%)] Loss: 0.3640
INFO:__main__:Train Epoch: 1 [480/6680 (7%)] Loss: 0.3871
INFO:__main__:Train Epoch: 1 [512/6680 (8%)] Loss: 0.4105
INFO:__main__:Train Epoch: 1 [544/6680 (8%)] Loss: 0.4332
INFO:__main__:Train Epoch: 1 [576/6680 (9%)] Loss: 0.4559
INFO:__main__:Train Epoch: 1 [608/6680 (9%)] Loss: 0.4787
INFO:__main__:Train Epoch: 1 [640/6680 (10%)] Loss: 0.5016
INFO:__main__:Train Epoch: 1 [672/6680 (10%)] Loss: 0.5250
INFO:__main__:Train Epoch: 1 [704/6680 (11%)] Loss: 0.5474
INFO:__main__:Train Epoch: 1 [736/6680 (11%)] Loss: 0.5706
INFO:__main__:Train Epoch: 1 [768/6680 (11%)] Loss: 0.5932
INFO:__main__:Train Epoch: 1 [800/6680 (12%)] Loss: 0.6157
INFO:__main__:Train Epoch: 1 [832/6680 (12%)] Loss: 0.6387
INFO:__main__:Train Epoch: 1 [864/6680 (13%)] Loss: 0.6602
INFO:__main__:Train Epoch: 1 [896/6680 (13%)] Loss: 0.6816
```

```
INFO:__main__:Train Epoch: 1 [928/6680 (14%)] Loss: 0.7036
INFO:__main__:Train Epoch: 1 [960/6680 (14%)] Loss: 0.7273
INFO:__main__:Train Epoch: 1 [992/6680 (15%)] Loss: 0.7493
INFO:__main__:Train Epoch: 1 [1024/6680 (15%)] Loss: 0.7716
INFO:__main__:Train Epoch: 1 [1056/6680 (16%)] Loss: 0.7913
INFO:__main__:Train Epoch: 1 [1088/6680 (16%)] Loss: 0.8133
INFO:__main__:Train Epoch: 1 [1120/6680 (17%)] Loss: 0.8341
INFO:__main__:Train Epoch: 1 [1152/6680 (17%)] Loss: 0.8552
INFO:__main__:Train Epoch: 1 [1184/6680 (18%)] Loss: 0.8763
INFO:__main__:Train Epoch: 1 [1216/6680 (18%)] Loss: 0.8987
INFO:__main__:Train Epoch: 1 [1248/6680 (19%)] Loss: 0.9200
INFO:__main__:Train Epoch: 1 [1280/6680 (19%)] Loss: 0.9410
INFO:__main__:Train Epoch: 1 [1312/6680 (20%)] Loss: 0.9623
INFO:__main__:Train Epoch: 1 [1344/6680 (20%)] Loss: 0.9838
INFO:__main__:Train Epoch: 1 [1376/6680 (21%)] Loss: 1.0048
INFO:__main__:Train Epoch: 1 [1408/6680 (21%)] Loss: 1.0260
INFO:__main__:Train Epoch: 1 [1440/6680 (22%)] Loss: 1.0474
INFO:__main__:Train Epoch: 1 [1472/6680 (22%)] Loss: 1.0669
INFO:__main__:Train Epoch: 1 [1504/6680 (23%)] Loss: 1.0876
INFO:__main__:Train Epoch: 1 [1536/6680 (23%)] Loss: 1.1078
INFO:__main__:Train Epoch: 1 [1568/6680 (23%)] Loss: 1.1271
INFO:__main__:Train Epoch: 1 [1600/6680 (24%)] Loss: 1.1469
INFO:__main__:Train Epoch: 1 [1632/6680 (24%)] Loss: 1.1668
INFO:__main__:Train Epoch: 1 [1664/6680 (25%)] Loss: 1.1870
INFO:__main__:Train Epoch: 1 [1696/6680 (25%)] Loss: 1.2070
INFO:__main__:Train Epoch: 1 [1728/6680 (26%)] Loss: 1.2275
INFO:__main__:Train Epoch: 1 [1760/6680 (26%)] Loss: 1.2461
INFO:__main__:Train Epoch: 1 [1792/6680 (27%)] Loss: 1.2659
INFO:__main__:Train Epoch: 1 [1824/6680 (27%)] Loss: 1.2852
INFO:__main__:Train Epoch: 1 [1856/6680 (28%)] Loss: 1.3041
INFO:__main__:Train Epoch: 1 [1888/6680 (28%)] Loss: 1.3234
INFO:__main__:Train Epoch: 1 [1920/6680 (29%)] Loss: 1.3429
INFO:__main__:Train Epoch: 1 [1952/6680 (29%)] Loss: 1.3628
INFO:__main__:Train Epoch: 1 [1984/6680 (30%)] Loss: 1.3823
INFO:__main__:Train Epoch: 1 [2016/6680 (30%)] Loss: 1.4006
INFO:__main__:Train Epoch: 1 [2048/6680 (31%)] Loss: 1.4193
INFO:__main__:Train Epoch: 1 [2080/6680 (31%)] Loss: 1.4372
INFO:__main__:Train Epoch: 1 [2112/6680 (32%)] Loss: 1.4549
INFO:__main__:Train Epoch: 1 [2144/6680 (32%)] Loss: 1.4739
INFO:__main__:Train Epoch: 1 [2176/6680 (33%)] Loss: 1.4921
INFO:__main__:Train Epoch: 1 [2208/6680 (33%)] Loss: 1.5103
INFO:__main__:Train Epoch: 1 [2240/6680 (34%)] Loss: 1.5273
INFO:__main__:Train Epoch: 1 [2272/6680 (34%)] Loss: 1.5456
INFO:__main__:Train Epoch: 1 [2304/6680 (34%)] Loss: 1.5640
INFO:__main__:Train Epoch: 1 [2336/6680 (35%)] Loss: 1.5806
INFO:__main__:Train Epoch: 1 [2368/6680 (35%)] Loss: 1.5958
INFO:__main__:Train Epoch: 1 [2400/6680 (36%)] Loss: 1.6139
INFO:__main__:Train Epoch: 1 [2432/6680 (36%)] Loss: 1.6303
INFO:__main__:Train Epoch: 1 [2464/6680 (37%)] Loss: 1.6465
INFO:__main__:Train Epoch: 1 [2496/6680 (37%)] Loss: 1.6621
INFO:__main__:Train Epoch: 1 [2528/6680 (38%)] Loss: 1.6798
INFO:__main__:Train Epoch: 1 [2560/6680 (38%)] Loss: 1.6969
INFO:__main__:Train Epoch: 1 [2592/6680 (39%)] Loss: 1.7132
INFO:__main__:Train Epoch: 1 [2624/6680 (39%)] Loss: 1.7301
INFO:__main__:Train Epoch: 1 [2656/6680 (40%)] Loss: 1.7469
INFO:__main__:Train Epoch: 1 [2688/6680 (40%)] Loss: 1.7635
INFO:__main__:Train Epoch: 1 [2720/6680 (41%)] Loss: 1.7790
INFO:__main__:Train Epoch: 1 [2752/6680 (41%)] Loss: 1.7975
INFO:__main__:Train Epoch: 1 [2784/6680 (42%)] Loss: 1.8142
2024-08-17 09:08:17,776 sagemaker-training-toolkit INFO Reporting training SUCCESS
INFO:__main__:Train Epoch: 1 [2816/6680 (42%)] Loss: 1.8290
INFO:__main__:Train Epoch: 1 [2848/6680 (43%)] Loss: 1.8450
INFO:__main__:Train Epoch: 1 [2880/6680 (43%)] Loss: 1.8596
INFO:__main__:Train Epoch: 1 [2912/6680 (44%)] Loss: 1.8752
INFO:__main__:Train Epoch: 1 [2944/6680 (44%)] Loss: 1.8904
INFO:__main__:Train Epoch: 1 [2976/6680 (45%)] Loss: 1.9038
INFO:__main__:Train Epoch: 1 [3008/6680 (45%)] Loss: 1.9185
INFO:__main__:Train Epoch: 1 [3040/6680 (46%)] Loss: 1.9325
INFO:__main__:Train Epoch: 1 [3072/6680 (46%)] Loss: 1.9463
INFO:__main__:Train Epoch: 1 [3104/6680 (46%)] Loss: 1.9613
INFO:__main__:Train Epoch: 1 [3136/6680 (47%)] Loss: 1.9751
INFO:__main__:Train Epoch: 1 [3168/6680 (47%)] Loss: 1.9889
INFO:__main__:Train Epoch: 1 [3200/6680 (48%)] Loss: 2.0030
INFO:__main__:Train Epoch: 1 [3232/6680 (48%)] Loss: 2.0172
INFO:__main__:Train Epoch: 1 [3264/6680 (49%)] Loss: 2.0319
INFO:__main__:Train Epoch: 1 [3296/6680 (49%)] Loss: 2.0440
INFO:__main__:Train Epoch: 1 [3328/6680 (50%)] Loss: 2.0586
INFO:__main__:Train Epoch: 1 [3360/6680 (50%)] Loss: 2.0742
INFO:__main__:Train Epoch: 1 [3392/6680 (51%)] Loss: 2.0872
INFO:__main__:Train Epoch: 1 [3424/6680 (51%)] Loss: 2.1015
INFO:__main__:Train Epoch: 1 [3456/6680 (52%)] Loss: 2.1146
INFO:__main__:Train Epoch: 1 [3488/6680 (52%)] Loss: 2.1260
INFO:__main__:Train Epoch: 1 [3520/6680 (53%)] Loss: 2.1383
INFO:__main__:Train Epoch: 1 [3552/6680 (53%)] Loss: 2.1512
```

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INFO: __main__:Train Epoch: 1 [3584/6680 (54%)] Loss: 2.1637
INFO: __main__:Train Epoch: 1 [3616/6680 (54%)] Loss: 2.1767
INFO: __main__:Train Epoch: 1 [3648/6680 (55%)] Loss: 2.1901
INFO: __main__:Train Epoch: 1 [3680/6680 (55%)] Loss: 2.2033
INFO: __main__:Train Epoch: 1 [3712/6680 (56%)] Loss: 2.2161
INFO: __main__:Train Epoch: 1 [3744/6680 (56%)] Loss: 2.2274
INFO: __main__:Train Epoch: 1 [3776/6680 (57%)] Loss: 2.2412
INFO: __main__:Train Epoch: 1 [3808/6680 (57%)] Loss: 2.2527
INFO: __main__:Train Epoch: 1 [3840/6680 (57%)] Loss: 2.2658
INFO: __main__:Train Epoch: 1 [3872/6680 (58%)] Loss: 2.2790
INFO: __main__:Train Epoch: 1 [3904/6680 (58%)] Loss: 2.2933
INFO: __main__:Train Epoch: 1 [3936/6680 (59%)] Loss: 2.3060
INFO: __main__:Train Epoch: 1 [3968/6680 (59%)] Loss: 2.3183
INFO: __main__:Train Epoch: 1 [4000/6680 (60%)] Loss: 2.3312
INFO: __main__:Train Epoch: 1 [4032/6680 (60%)] Loss: 2.3423
INFO: __main__:Train Epoch: 1 [4064/6680 (61%)] Loss: 2.3549
INFO: __main__:Train Epoch: 1 [4096/6680 (61%)] Loss: 2.3678
INFO: __main__:Train Epoch: 1 [4128/6680 (62%)] Loss: 2.3816
INFO: __main__:Train Epoch: 1 [4160/6680 (62%)] Loss: 2.3920
INFO: __main__:Train Epoch: 1 [4192/6680 (63%)] Loss: 2.4040
INFO: __main__:Train Epoch: 1 [4224/6680 (63%)] Loss: 2.4145
INFO: __main__:Train Epoch: 1 [4256/6680 (64%)] Loss: 2.4252
INFO: __main__:Train Epoch: 1 [4288/6680 (64%)] Loss: 2.4365
INFO: __main__:Train Epoch: 1 [4320/6680 (65%)] Loss: 2.4478
INFO: __main__:Train Epoch: 1 [4352/6680 (65%)] Loss: 2.4586
INFO: __main__:Train Epoch: 1 [4384/6680 (66%)] Loss: 2.4707
INFO: __main__:Train Epoch: 1 [4416/6680 (66%)] Loss: 2.4820
INFO: __main__:Train Epoch: 1 [4448/6680 (67%)] Loss: 2.4933
INFO: __main__:Train Epoch: 1 [4480/6680 (67%)] Loss: 2.5036
INFO: __main__:Train Epoch: 1 [4512/6680 (68%)] Loss: 2.5137
INFO: __main__:Train Epoch: 1 [4544/6680 (68%)] Loss: 2.5239
INFO: __main__:Train Epoch: 1 [4576/6680 (69%)] Loss: 2.5352
INFO: __main__:Train Epoch: 1 [4608/6680 (69%)] Loss: 2.5458
INFO: __main__:Train Epoch: 1 [4640/6680 (69%)] Loss: 2.5569
INFO: __main__:Train Epoch: 1 [4672/6680 (70%)] Loss: 2.5692
INFO: __main__:Train Epoch: 1 [4704/6680 (70%)] Loss: 2.5809
INFO: __main__:Train Epoch: 1 [4736/6680 (71%)] Loss: 2.5914
INFO: __main__:Train Epoch: 1 [4768/6680 (71%)] Loss: 2.6026
INFO: __main__:Train Epoch: 1 [4800/6680 (72%)] Loss: 2.6136
INFO: __main__:Train Epoch: 1 [4832/6680 (72%)] Loss: 2.6238
INFO: __main__:Train Epoch: 1 [4864/6680 (73%)] Loss: 2.6355
INFO: __main__:Train Epoch: 1 [4896/6680 (73%)] Loss: 2.6462
INFO: __main__:Train Epoch: 1 [4928/6680 (74%)] Loss: 2.6560
INFO: __main__:Train Epoch: 1 [4960/6680 (74%)] Loss: 2.6661
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INFO: __main__:Train Epoch: 1 [5024/6680 (75%)] Loss: 2.6878
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INFO: __main__:Train Epoch: 1 [5216/6680 (78%)] Loss: 2.7490
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INFO: __main__:Train Epoch: 1 [5280/6680 (79%)] Loss: 2.7682
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INFO: __main__:Train Epoch: 1 [5344/6680 (80%)] Loss: 2.7874
INFO: __main__:Train Epoch: 1 [5376/6680 (80%)] Loss: 2.7967
INFO: __main__:Train Epoch: 1 [5408/6680 (81%)] Loss: 2.8064
INFO: __main__:Train Epoch: 1 [5440/6680 (81%)] Loss: 2.8161
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INFO: __main__:Train Epoch: 1 [5568/6680 (83%)] Loss: 2.8520
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INFO: __main__:Train Epoch: 1 [5920/6680 (89%)] Loss: 2.9593
INFO: __main__:Train Epoch: 1 [5952/6680 (89%)] Loss: 2.9677
INFO: __main__:Train Epoch: 1 [5984/6680 (90%)] Loss: 2.9785
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INFO: __main__:Train Epoch: 1 [6048/6680 (91%)] Loss: 2.9986
INFO: __main__:Train Epoch: 1 [6080/6680 (91%)] Loss: 3.0076
INFO: __main__:Train Epoch: 1 [6112/6680 (91%)] Loss: 3.0170
INFO: __main__:Train Epoch: 1 [6144/6680 (92%)] Loss: 3.0269
INFO: __main__:Train Epoch: 1 [6176/6680 (92%)] Loss: 3.0352
INFO: __main__:Train Epoch: 1 [6208/6680 (93%)] Loss: 3.0439
INFO: __main__:Train Epoch: 1 [6240/6680 (93%)] Loss: 3.0530
```

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INFO:__main__:Train Epoch: 1 [6336/6680 (95%)] Loss: 3.0795
INFO:__main__:Train Epoch: 1 [6368/6680 (95%)] Loss: 3.0873
INFO:__main__:Train Epoch: 1 [6400/6680 (96%)] Loss: 3.0950
INFO:__main__:Train Epoch: 1 [6432/6680 (96%)] Loss: 3.1021
INFO:__main__:Train Epoch: 1 [6464/6680 (97%)] Loss: 3.1104
INFO:__main__:Train Epoch: 1 [6496/6680 (97%)] Loss: 3.1197
INFO:__main__:Train Epoch: 1 [6528/6680 (98%)] Loss: 3.1271
INFO:__main__:Train Epoch: 1 [6560/6680 (98%)] Loss: 3.1362
INFO:__main__:Train Epoch: 1 [6592/6680 (99%)] Loss: 3.1438
INFO:__main__:Train Epoch: 1 [6624/6680 (99%)] Loss: 3.1539
INFO:__main__:Train Epoch: 1 [6656/6680 (100%)] Loss: 3.1612
INFO:__main__:Train Epoch: 1 [5016/6680 (75%)] Loss: 3.1707
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INFO:__main__:Validate Epoch: 1 [64/835 (8%)] Loss: 0.1129
INFO:__main__:Validate Epoch: 1 [96/835 (11%)] Loss: 0.1757
INFO:__main__:Validate Epoch: 1 [128/835 (15%)] Loss: 0.2188
INFO:__main__:Validate Epoch: 1 [160/835 (19%)] Loss: 0.2462
INFO:__main__:Validate Epoch: 1 [192/835 (23%)] Loss: 0.2973
INFO:__main__:Validate Epoch: 1 [224/835 (27%)] Loss: 0.3589
INFO:__main__:Validate Epoch: 1 [256/835 (31%)] Loss: 0.4376
INFO:__main__:Validate Epoch: 1 [288/835 (34%)] Loss: 0.4915
INFO:__main__:Validate Epoch: 1 [320/835 (38%)] Loss: 0.5439
INFO:__main__:Validate Epoch: 1 [352/835 (42%)] Loss: 0.6155
INFO:__main__:Validate Epoch: 1 [384/835 (46%)] Loss: 0.6678
INFO:__main__:Validate Epoch: 1 [416/835 (50%)] Loss: 0.7373
INFO:__main__:Validate Epoch: 1 [448/835 (54%)] Loss: 0.7922
INFO:__main__:Validate Epoch: 1 [480/835 (57%)] Loss: 0.8703
INFO:__main__:Validate Epoch: 1 [512/835 (61%)] Loss: 0.9590
INFO:__main__:Validate Epoch: 1 [544/835 (65%)] Loss: 1.0294
INFO:__main__:Validate Epoch: 1 [576/835 (69%)] Loss: 1.0913
INFO:__main__:Validate Epoch: 1 [608/835 (73%)] Loss: 1.1495
INFO:__main__:Validate Epoch: 1 [640/835 (77%)] Loss: 1.1979
INFO:__main__:Validate Epoch: 1 [672/835 (80%)] Loss: 1.2764
INFO:__main__:Validate Epoch: 1 [704/835 (84%)] Loss: 1.3423
INFO:__main__:Validate Epoch: 1 [736/835 (88%)] Loss: 1.4149
INFO:__main__:Validate Epoch: 1 [768/835 (92%)] Loss: 1.4816
INFO:__main__:Validate Epoch: 1 [800/835 (96%)] Loss: 1.5521
INFO:__main__:Validate Epoch: 1 [832/835 (100%)] Loss: 1.6225
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INFO:__main__:Train Epoch: 2 [1344/6680 (20%)] Loss: 0.3347
INFO:__main__:Train Epoch: 2 [1376/6680 (21%)] Loss: 0.3435
```



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INFO:__main__:Train Epoch: 2 [1472/6680 (22%)] Loss: 0.3643
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INFO:__main__:Train Epoch: 2 [3968/6680 (59%)] Loss: 0.8867
INFO:__main__:Train Epoch: 2 [4000/6680 (60%)] Loss: 0.8928
INFO:__main__:Train Epoch: 2 [4032/6680 (60%)] Loss: 0.8977
INFO:__main__:Train Epoch: 2 [4064/6680 (61%)] Loss: 0.9057
```

```
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INFO:__main__:Train Epoch: 2 [4832/6680 (72%)] Loss: 1.0485
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INFO:__main__:Train Epoch: 2 [6368/6680 (95%)] Loss: 1.3206
INFO:__main__:Train Epoch: 2 [6400/6680 (96%)] Loss: 1.3274
INFO:__main__:Train Epoch: 2 [6432/6680 (96%)] Loss: 1.3350
INFO:__main__:Train Epoch: 2 [6464/6680 (97%)] Loss: 1.3411
INFO:__main__:Train Epoch: 2 [6496/6680 (97%)] Loss: 1.3466
INFO:__main__:Train Epoch: 2 [6528/6680 (98%)] Loss: 1.3528
INFO:__main__:Train Epoch: 2 [6560/6680 (98%)] Loss: 1.3596
INFO:__main__:Train Epoch: 2 [6592/6680 (99%)] Loss: 1.3658
INFO:__main__:Train Epoch: 2 [6624/6680 (99%)] Loss: 1.3717
INFO:__main__:Train Epoch: 2 [6656/6680 (100%)] Loss: 1.3774
INFO:__main__:Train Epoch: 2 [5016/6680 (75%)] Loss: 1.3836
INFO:__main__:Validate Epoch: 2 [32/835 (4%)] Loss: 0.0314
INFO:__main__:Validate Epoch: 2 [64/835 (8%)] Loss: 0.0694
```



```
INFO:__main__:Validate Epoch: 2 [96/835 (11%)] Loss: 0.1137
INFO:__main__:Validate Epoch: 2 [128/835 (15%)] Loss: 0.1368
INFO:__main__:Validate Epoch: 2 [160/835 (19%)] Loss: 0.1550
INFO:__main__:Validate Epoch: 2 [192/835 (23%)] Loss: 0.1900
INFO:__main__:Validate Epoch: 2 [224/835 (27%)] Loss: 0.2312
INFO:__main__:Validate Epoch: 2 [256/835 (31%)] Loss: 0.2793
INFO:__main__:Validate Epoch: 2 [288/835 (34%)] Loss: 0.3161
INFO:__main__:Validate Epoch: 2 [320/835 (38%)] Loss: 0.3580
INFO:__main__:Validate Epoch: 2 [352/835 (42%)] Loss: 0.3994
INFO:__main__:Validate Epoch: 2 [384/835 (46%)] Loss: 0.4353
INFO:__main__:Validate Epoch: 2 [416/835 (50%)] Loss: 0.4747
INFO:__main__:Validate Epoch: 2 [448/835 (54%)] Loss: 0.5129
INFO:__main__:Validate Epoch: 2 [480/835 (57%)] Loss: 0.5616
INFO:__main__:Validate Epoch: 2 [512/835 (61%)] Loss: 0.6188
INFO:__main__:Validate Epoch: 2 [544/835 (65%)] Loss: 0.6719
INFO:__main__:Validate Epoch: 2 [576/835 (69%)] Loss: 0.7215
INFO:__main__:Validate Epoch: 2 [608/835 (73%)] Loss: 0.7557
INFO:__main__:Validate Epoch: 2 [640/835 (77%)] Loss: 0.7786
INFO:__main__:Validate Epoch: 2 [672/835 (80%)] Loss: 0.8286
INFO:__main__:Validate Epoch: 2 [704/835 (84%)] Loss: 0.8785
INFO:__main__:Validate Epoch: 2 [736/835 (88%)] Loss: 0.9175
INFO:__main__:Validate Epoch: 2 [768/835 (92%)] Loss: 0.9613
INFO:__main__:Validate Epoch: 2 [800/835 (96%)] Loss: 1.0113
INFO:__main__:Validate Epoch: 2 [832/835 (100%)] Loss: 1.0573
INFO:__main__:Validate Epoch: 2 [81/835 (10%)] Loss: 1.0883
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INFO:__main__:Train Epoch: 3 [96/6680 (1%)] Loss: 0.0170
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INFO:__main__:Train Epoch: 3 [160/6680 (2%)] Loss: 0.0272
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INFO:__main__:Train Epoch: 3 [224/6680 (3%)] Loss: 0.0385
INFO:__main__:Train Epoch: 3 [256/6680 (4%)] Loss: 0.0438
INFO:__main__:Train Epoch: 3 [288/6680 (4%)] Loss: 0.0476
INFO:__main__:Train Epoch: 3 [320/6680 (5%)] Loss: 0.0522
INFO:__main__:Train Epoch: 3 [352/6680 (5%)] Loss: 0.0567
INFO:__main__:Train Epoch: 3 [384/6680 (6%)] Loss: 0.0620
INFO:__main__:Train Epoch: 3 [416/6680 (6%)] Loss: 0.0686
INFO:__main__:Train Epoch: 3 [448/6680 (7%)] Loss: 0.0733
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INFO:__main__:Train Epoch: 3 [576/6680 (9%)] Loss: 0.0955
INFO:__main__:Train Epoch: 3 [608/6680 (9%)] Loss: 0.1010
INFO:__main__:Train Epoch: 3 [640/6680 (10%)] Loss: 0.1059
INFO:__main__:Train Epoch: 3 [672/6680 (10%)] Loss: 0.1113
INFO:__main__:Train Epoch: 3 [704/6680 (11%)] Loss: 0.1151
INFO:__main__:Train Epoch: 3 [736/6680 (11%)] Loss: 0.1209
INFO:__main__:Train Epoch: 3 [768/6680 (11%)] Loss: 0.1255
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INFO:__main__:Train Epoch: 3 [832/6680 (12%)] Loss: 0.1353
INFO:__main__:Train Epoch: 3 [864/6680 (13%)] Loss: 0.1412
INFO:__main__:Train Epoch: 3 [896/6680 (13%)] Loss: 0.1460
INFO:__main__:Train Epoch: 3 [928/6680 (14%)] Loss: 0.1520
INFO:__main__:Train Epoch: 3 [960/6680 (14%)] Loss: 0.1561
INFO:__main__:Train Epoch: 3 [992/6680 (15%)] Loss: 0.1606
INFO:__main__:Train Epoch: 3 [1024/6680 (15%)] Loss: 0.1672
INFO:__main__:Train Epoch: 3 [1056/6680 (16%)] Loss: 0.1732
INFO:__main__:Train Epoch: 3 [1088/6680 (16%)] Loss: 0.1792
INFO:__main__:Train Epoch: 3 [1120/6680 (17%)] Loss: 0.1849
INFO:__main__:Train Epoch: 3 [1152/6680 (17%)] Loss: 0.1895
INFO:__main__:Train Epoch: 3 [1184/6680 (18%)] Loss: 0.1938
INFO:__main__:Train Epoch: 3 [1216/6680 (18%)] Loss: 0.2005
INFO:__main__:Train Epoch: 3 [1248/6680 (19%)] Loss: 0.2050
INFO:__main__:Train Epoch: 3 [1280/6680 (19%)] Loss: 0.2094
INFO:__main__:Train Epoch: 3 [1312/6680 (20%)] Loss: 0.2150
INFO:__main__:Train Epoch: 3 [1344/6680 (20%)] Loss: 0.2201
INFO:__main__:Train Epoch: 3 [1376/6680 (21%)] Loss: 0.2259
INFO:__main__:Train Epoch: 3 [1408/6680 (21%)] Loss: 0.2302
INFO:__main__:Train Epoch: 3 [1440/6680 (22%)] Loss: 0.2361
INFO:__main__:Train Epoch: 3 [1472/6680 (22%)] Loss: 0.2406
INFO:__main__:Train Epoch: 3 [1504/6680 (23%)] Loss: 0.2456
INFO:__main__:Train Epoch: 3 [1536/6680 (23%)] Loss: 0.2492
INFO:__main__:Train Epoch: 3 [1568/6680 (23%)] Loss: 0.2542
INFO:__main__:Train Epoch: 3 [1600/6680 (24%)] Loss: 0.2585
INFO:__main__:Train Epoch: 3 [1632/6680 (24%)] Loss: 0.2650
INFO:__main__:Train Epoch: 3 [1664/6680 (25%)] Loss: 0.2708
INFO:__main__:Train Epoch: 3 [1696/6680 (25%)] Loss: 0.2748
INFO:__main__:Train Epoch: 3 [1728/6680 (26%)] Loss: 0.2797
INFO:__main__:Train Epoch: 3 [1760/6680 (26%)] Loss: 0.2848
INFO:__main__:Train Epoch: 3 [1792/6680 (27%)] Loss: 0.2897
INFO:__main__:Train Epoch: 3 [1824/6680 (27%)] Loss: 0.2943
INFO:__main__:Train Epoch: 3 [1856/6680 (28%)] Loss: 0.2993
INFO:__main__:Train Epoch: 3 [1888/6680 (28%)] Loss: 0.3031
```

```
INFO:__main__:Train Epoch: 3 [1920/6680 (29%)] Loss: 0.3078
INFO:__main__:Train Epoch: 3 [1952/6680 (29%)] Loss: 0.3119
INFO:__main__:Train Epoch: 3 [1984/6680 (30%)] Loss: 0.3171
INFO:__main__:Train Epoch: 3 [2016/6680 (30%)] Loss: 0.3224
INFO:__main__:Train Epoch: 3 [2048/6680 (31%)] Loss: 0.3274
INFO:__main__:Train Epoch: 3 [2080/6680 (31%)] Loss: 0.3323
INFO:__main__:Train Epoch: 3 [2112/6680 (32%)] Loss: 0.3356
INFO:__main__:Train Epoch: 3 [2144/6680 (32%)] Loss: 0.3416
INFO:__main__:Train Epoch: 3 [2176/6680 (33%)] Loss: 0.3452
INFO:__main__:Train Epoch: 3 [2208/6680 (33%)] Loss: 0.3492
INFO:__main__:Train Epoch: 3 [2240/6680 (34%)] Loss: 0.3551
INFO:__main__:Train Epoch: 3 [2272/6680 (34%)] Loss: 0.3607
INFO:__main__:Train Epoch: 3 [2304/6680 (34%)] Loss: 0.3644
INFO:__main__:Train Epoch: 3 [2336/6680 (35%)] Loss: 0.3694
INFO:__main__:Train Epoch: 3 [2368/6680 (35%)] Loss: 0.3735
INFO:__main__:Train Epoch: 3 [2400/6680 (36%)] Loss: 0.3785
INFO:__main__:Train Epoch: 3 [2432/6680 (36%)] Loss: 0.3844
INFO:__main__:Train Epoch: 3 [2464/6680 (37%)] Loss: 0.3891
INFO:__main__:Train Epoch: 3 [2496/6680 (37%)] Loss: 0.3975
INFO:__main__:Train Epoch: 3 [2528/6680 (38%)] Loss: 0.4026
INFO:__main__:Train Epoch: 3 [2560/6680 (38%)] Loss: 0.4078
INFO:__main__:Train Epoch: 3 [2592/6680 (39%)] Loss: 0.4121
INFO:__main__:Train Epoch: 3 [2624/6680 (39%)] Loss: 0.4158
INFO:__main__:Train Epoch: 3 [2656/6680 (40%)] Loss: 0.4214
INFO:__main__:Train Epoch: 3 [2688/6680 (40%)] Loss: 0.4273
INFO:__main__:Train Epoch: 3 [2720/6680 (41%)] Loss: 0.4329
INFO:__main__:Train Epoch: 3 [2752/6680 (41%)] Loss: 0.4385
INFO:__main__:Train Epoch: 3 [2784/6680 (42%)] Loss: 0.4455
INFO:__main__:Train Epoch: 3 [2816/6680 (42%)] Loss: 0.4508
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INFO:__main__:Train Epoch: 3 [2880/6680 (43%)] Loss: 0.4620
INFO:__main__:Train Epoch: 3 [2912/6680 (44%)] Loss: 0.4667
INFO:__main__:Train Epoch: 3 [2944/6680 (44%)] Loss: 0.4717
INFO:__main__:Train Epoch: 3 [2976/6680 (45%)] Loss: 0.4775
INFO:__main__:Train Epoch: 3 [3008/6680 (45%)] Loss: 0.4825
INFO:__main__:Train Epoch: 3 [3040/6680 (46%)] Loss: 0.4866
INFO:__main__:Train Epoch: 3 [3072/6680 (46%)] Loss: 0.4920
INFO:__main__:Train Epoch: 3 [3104/6680 (46%)] Loss: 0.4968
INFO:__main__:Train Epoch: 3 [3136/6680 (47%)] Loss: 0.5002
INFO:__main__:Train Epoch: 3 [3168/6680 (47%)] Loss: 0.5067
INFO:__main__:Train Epoch: 3 [3200/6680 (48%)] Loss: 0.5131
INFO:__main__:Train Epoch: 3 [3232/6680 (48%)] Loss: 0.5173
INFO:__main__:Train Epoch: 3 [3264/6680 (49%)] Loss: 0.5232
INFO:__main__:Train Epoch: 3 [3296/6680 (49%)] Loss: 0.5270
INFO:__main__:Train Epoch: 3 [3328/6680 (50%)] Loss: 0.5319
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INFO:__main__:Train Epoch: 3 [3392/6680 (51%)] Loss: 0.5402
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INFO:__main__:Train Epoch: 3 [3584/6680 (54%)] Loss: 0.5700
INFO:__main__:Train Epoch: 3 [3616/6680 (54%)] Loss: 0.5748
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INFO:__main__:Train Epoch: 3 [3712/6680 (56%)] Loss: 0.5903
INFO:__main__:Train Epoch: 3 [3744/6680 (56%)] Loss: 0.5959
INFO:__main__:Train Epoch: 3 [3776/6680 (57%)] Loss: 0.5999
INFO:__main__:Train Epoch: 3 [3808/6680 (57%)] Loss: 0.6039
INFO:__main__:Train Epoch: 3 [3840/6680 (57%)] Loss: 0.6093
INFO:__main__:Train Epoch: 3 [3872/6680 (58%)] Loss: 0.6143
INFO:__main__:Train Epoch: 3 [3904/6680 (58%)] Loss: 0.6193
INFO:__main__:Train Epoch: 3 [3936/6680 (59%)] Loss: 0.6244
INFO:__main__:Train Epoch: 3 [3968/6680 (59%)] Loss: 0.6280
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INFO:__main__:Train Epoch: 3 [4192/6680 (63%)] Loss: 0.6588
INFO:__main__:Train Epoch: 3 [4224/6680 (63%)] Loss: 0.6630
INFO:__main__:Train Epoch: 3 [4256/6680 (64%)] Loss: 0.6684
INFO:__main__:Train Epoch: 3 [4288/6680 (64%)] Loss: 0.6729
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INFO:__main__:Train Epoch: 3 [4352/6680 (65%)] Loss: 0.6819
INFO:__main__:Train Epoch: 3 [4384/6680 (66%)] Loss: 0.6861
INFO:__main__:Train Epoch: 3 [4416/6680 (66%)] Loss: 0.6920
INFO:__main__:Train Epoch: 3 [4448/6680 (67%)] Loss: 0.6969
INFO:__main__:Train Epoch: 3 [4480/6680 (67%)] Loss: 0.7023
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INFO:__main__:Train Epoch: 3 [4544/6680 (68%)] Loss: 0.7118
INFO:__main__:Train Epoch: 3 [4576/6680 (69%)] Loss: 0.7160
```

```
INFO:__main__:Train Epoch: 3 [4608/6680 (69%)] Loss: 0.7193
INFO:__main__:Train Epoch: 3 [4640/6680 (69%)] Loss: 0.7250
INFO:__main__:Train Epoch: 3 [4672/6680 (70%)] Loss: 0.7295
INFO:__main__:Train Epoch: 3 [4704/6680 (70%)] Loss: 0.7334
INFO:__main__:Train Epoch: 3 [4736/6680 (71%)] Loss: 0.7385
INFO:__main__:Train Epoch: 3 [4768/6680 (71%)] Loss: 0.7424
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INFO:__main__:Train Epoch: 3 [4832/6680 (72%)] Loss: 0.7508
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INFO:__main__:Train Epoch: 3 [4896/6680 (73%)] Loss: 0.7596
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INFO:__main__:Train Epoch: 3 [4960/6680 (74%)] Loss: 0.7706
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INFO:__main__:Train Epoch: 3 [5024/6680 (75%)] Loss: 0.7792
INFO:__main__:Train Epoch: 3 [5056/6680 (76%)] Loss: 0.7833
INFO:__main__:Train Epoch: 3 [5088/6680 (76%)] Loss: 0.7874
INFO:__main__:Train Epoch: 3 [5120/6680 (77%)] Loss: 0.7922
INFO:__main__:Train Epoch: 3 [5152/6680 (77%)] Loss: 0.7976
INFO:__main__:Train Epoch: 3 [5184/6680 (78%)] Loss: 0.8015
INFO:__main__:Train Epoch: 3 [5216/6680 (78%)] Loss: 0.8052
INFO:__main__:Train Epoch: 3 [5248/6680 (79%)] Loss: 0.8091
INFO:__main__:Train Epoch: 3 [5280/6680 (79%)] Loss: 0.8148
INFO:__main__:Train Epoch: 3 [5312/6680 (80%)] Loss: 0.8193
INFO:__main__:Train Epoch: 3 [5344/6680 (80%)] Loss: 0.8237
INFO:__main__:Train Epoch: 3 [5376/6680 (80%)] Loss: 0.8296
INFO:__main__:Train Epoch: 3 [5408/6680 (81%)] Loss: 0.8338
INFO:__main__:Train Epoch: 3 [5440/6680 (81%)] Loss: 0.8383
INFO:__main__:Train Epoch: 3 [5472/6680 (82%)] Loss: 0.8450
INFO:__main__:Train Epoch: 3 [5504/6680 (82%)] Loss: 0.8490
INFO:__main__:Train Epoch: 3 [5536/6680 (83%)] Loss: 0.8543
INFO:__main__:Train Epoch: 3 [5568/6680 (83%)] Loss: 0.8603
INFO:__main__:Train Epoch: 3 [5600/6680 (84%)] Loss: 0.8640
INFO:__main__:Train Epoch: 3 [5632/6680 (84%)] Loss: 0.8677
INFO:__main__:Train Epoch: 3 [5664/6680 (85%)] Loss: 0.8744
INFO:__main__:Train Epoch: 3 [5696/6680 (85%)] Loss: 0.8784
INFO:__main__:Train Epoch: 3 [5728/6680 (86%)] Loss: 0.8839
INFO:__main__:Train Epoch: 3 [5760/6680 (86%)] Loss: 0.8889
INFO:__main__:Train Epoch: 3 [5792/6680 (87%)] Loss: 0.8931
INFO:__main__:Train Epoch: 3 [5824/6680 (87%)] Loss: 0.8973
INFO:__main__:Train Epoch: 3 [5856/6680 (88%)] Loss: 0.9032
INFO:__main__:Train Epoch: 3 [5888/6680 (88%)] Loss: 0.9078
INFO:__main__:Train Epoch: 3 [5920/6680 (89%)] Loss: 0.9119
INFO:__main__:Train Epoch: 3 [5952/6680 (89%)] Loss: 0.9178
INFO:__main__:Train Epoch: 3 [5984/6680 (90%)] Loss: 0.9235
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INFO:__main__:Train Epoch: 3 [6080/6680 (91%)] Loss: 0.9363
INFO:__main__:Train Epoch: 3 [6112/6680 (91%)] Loss: 0.9416
INFO:__main__:Train Epoch: 3 [6144/6680 (92%)] Loss: 0.9463
INFO:__main__:Train Epoch: 3 [6176/6680 (92%)] Loss: 0.9499
INFO:__main__:Train Epoch: 3 [6208/6680 (93%)] Loss: 0.9560
INFO:__main__:Train Epoch: 3 [6240/6680 (93%)] Loss: 0.9605
INFO:__main__:Train Epoch: 3 [6272/6680 (94%)] Loss: 0.9644
INFO:__main__:Train Epoch: 3 [6304/6680 (94%)] Loss: 0.9690
INFO:__main__:Train Epoch: 3 [6336/6680 (95%)] Loss: 0.9728
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INFO:__main__:Train Epoch: 3 [6400/6680 (96%)] Loss: 0.9821
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INFO:__main__:Train Epoch: 3 [6464/6680 (97%)] Loss: 0.9917
INFO:__main__:Train Epoch: 3 [6496/6680 (97%)] Loss: 0.9945
INFO:__main__:Train Epoch: 3 [6528/6680 (98%)] Loss: 0.9988
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INFO:__main__:Train Epoch: 3 [6592/6680 (99%)] Loss: 1.0107
INFO:__main__:Train Epoch: 3 [6624/6680 (99%)] Loss: 1.0157
INFO:__main__:Train Epoch: 3 [6656/6680 (100%)] Loss: 1.0216
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INFO:__main__:Validate Epoch: 3 [32/835 (4%)] Loss: 0.0215
INFO:__main__:Validate Epoch: 3 [64/835 (8%)] Loss: 0.0516
INFO:__main__:Validate Epoch: 3 [96/835 (11%)] Loss: 0.0863
INFO:__main__:Validate Epoch: 3 [128/835 (15%)] Loss: 0.1090
INFO:__main__:Validate Epoch: 3 [160/835 (19%)] Loss: 0.1285
INFO:__main__:Validate Epoch: 3 [192/835 (23%)] Loss: 0.1541
INFO:__main__:Validate Epoch: 3 [224/835 (27%)] Loss: 0.1928
INFO:__main__:Validate Epoch: 3 [256/835 (31%)] Loss: 0.2351
INFO:__main__:Validate Epoch: 3 [288/835 (34%)] Loss: 0.2729
INFO:__main__:Validate Epoch: 3 [320/835 (38%)] Loss: 0.3042
INFO:__main__:Validate Epoch: 3 [352/835 (42%)] Loss: 0.3423
INFO:__main__:Validate Epoch: 3 [384/835 (46%)] Loss: 0.3756
INFO:__main__:Validate Epoch: 3 [416/835 (50%)] Loss: 0.4185
INFO:__main__:Validate Epoch: 3 [448/835 (54%)] Loss: 0.4478
INFO:__main__:Validate Epoch: 3 [480/835 (57%)] Loss: 0.4898
INFO:__main__:Validate Epoch: 3 [512/835 (61%)] Loss: 0.5372
INFO:__main__:Validate Epoch: 3 [544/835 (65%)] Loss: 0.5826
INFO:__main__:Validate Epoch: 3 [576/835 (69%)] Loss: 0.6282
```

```
INFO:__main__:Validate Epoch: 3 [608/835 (73%)] Loss: 0.6537
INFO:__main__:Validate Epoch: 3 [640/835 (77%)] Loss: 0.6807
INFO:__main__:Validate Epoch: 3 [672/835 (80%)] Loss: 0.7214
INFO:__main__:Validate Epoch: 3 [704/835 (84%)] Loss: 0.7584
INFO:__main__:Validate Epoch: 3 [736/835 (88%)] Loss: 0.7947
INFO:__main__:Validate Epoch: 3 [768/835 (92%)] Loss: 0.8330
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INFO:__main__:Train Epoch: 4 [224/6680 (3%)] Loss: 0.0316
INFO:__main__:Train Epoch: 4 [256/6680 (4%)] Loss: 0.0359
INFO:__main__:Train Epoch: 4 [288/6680 (4%)] Loss: 0.0399
INFO:__main__:Train Epoch: 4 [320/6680 (5%)] Loss: 0.0429
INFO:__main__:Train Epoch: 4 [352/6680 (5%)] Loss: 0.0473
INFO:__main__:Train Epoch: 4 [384/6680 (6%)] Loss: 0.0526
INFO:__main__:Train Epoch: 4 [416/6680 (6%)] Loss: 0.0566
INFO:__main__:Train Epoch: 4 [448/6680 (7%)] Loss: 0.0611
INFO:__main__:Train Epoch: 4 [480/6680 (7%)] Loss: 0.0650
INFO:__main__:Train Epoch: 4 [512/6680 (8%)] Loss: 0.0700
INFO:__main__:Train Epoch: 4 [544/6680 (8%)] Loss: 0.0729
INFO:__main__:Train Epoch: 4 [576/6680 (9%)] Loss: 0.0768
INFO:__main__:Train Epoch: 4 [608/6680 (9%)] Loss: 0.0807
INFO:__main__:Train Epoch: 4 [640/6680 (10%)] Loss: 0.0856
INFO:__main__:Train Epoch: 4 [672/6680 (10%)] Loss: 0.0909
INFO:__main__:Train Epoch: 4 [704/6680 (11%)] Loss: 0.0943
INFO:__main__:Train Epoch: 4 [736/6680 (11%)] Loss: 0.0998
INFO:__main__:Train Epoch: 4 [768/6680 (11%)] Loss: 0.1033
INFO:__main__:Train Epoch: 4 [800/6680 (12%)] Loss: 0.1086
INFO:__main__:Train Epoch: 4 [832/6680 (12%)] Loss: 0.1135
INFO:__main__:Train Epoch: 4 [864/6680 (13%)] Loss: 0.1179
INFO:__main__:Train Epoch: 4 [896/6680 (13%)] Loss: 0.1216
INFO:__main__:Train Epoch: 4 [928/6680 (14%)] Loss: 0.1266
INFO:__main__:Train Epoch: 4 [960/6680 (14%)] Loss: 0.1326
INFO:__main__:Train Epoch: 4 [992/6680 (15%)] Loss: 0.1368
INFO:__main__:Train Epoch: 4 [1024/6680 (15%)] Loss: 0.1419
INFO:__main__:Train Epoch: 4 [1056/6680 (16%)] Loss: 0.1457
INFO:__main__:Train Epoch: 4 [1088/6680 (16%)] Loss: 0.1490
INFO:__main__:Train Epoch: 4 [1120/6680 (17%)] Loss: 0.1528
INFO:__main__:Train Epoch: 4 [1152/6680 (17%)] Loss: 0.1576
INFO:__main__:Train Epoch: 4 [1184/6680 (18%)] Loss: 0.1617
INFO:__main__:Train Epoch: 4 [1216/6680 (18%)] Loss: 0.1668
INFO:__main__:Train Epoch: 4 [1248/6680 (19%)] Loss: 0.1712
INFO:__main__:Train Epoch: 4 [1280/6680 (19%)] Loss: 0.1749
INFO:__main__:Train Epoch: 4 [1312/6680 (20%)] Loss: 0.1782
INFO:__main__:Train Epoch: 4 [1344/6680 (20%)] Loss: 0.1830
INFO:__main__:Train Epoch: 4 [1376/6680 (21%)] Loss: 0.1882
INFO:__main__:Train Epoch: 4 [1408/6680 (21%)] Loss: 0.1933
INFO:__main__:Train Epoch: 4 [1440/6680 (22%)] Loss: 0.1964
INFO:__main__:Train Epoch: 4 [1472/6680 (22%)] Loss: 0.2019
INFO:__main__:Train Epoch: 4 [1504/6680 (23%)] Loss: 0.2071
INFO:__main__:Train Epoch: 4 [1536/6680 (23%)] Loss: 0.2123
INFO:__main__:Train Epoch: 4 [1568/6680 (23%)] Loss: 0.2174
INFO:__main__:Train Epoch: 4 [1600/6680 (24%)] Loss: 0.2219
INFO:__main__:Train Epoch: 4 [1632/6680 (24%)] Loss: 0.2267
INFO:__main__:Train Epoch: 4 [1664/6680 (25%)] Loss: 0.2307
INFO:__main__:Train Epoch: 4 [1696/6680 (25%)] Loss: 0.2350
INFO:__main__:Train Epoch: 4 [1728/6680 (26%)] Loss: 0.2410
INFO:__main__:Train Epoch: 4 [1760/6680 (26%)] Loss: 0.2470
INFO:__main__:Train Epoch: 4 [1792/6680 (27%)] Loss: 0.2512
INFO:__main__:Train Epoch: 4 [1824/6680 (27%)] Loss: 0.2552
INFO:__main__:Train Epoch: 4 [1856/6680 (28%)] Loss: 0.2590
INFO:__main__:Train Epoch: 4 [1888/6680 (28%)] Loss: 0.2633
INFO:__main__:Train Epoch: 4 [1920/6680 (29%)] Loss: 0.2675
INFO:__main__:Train Epoch: 4 [1952/6680 (29%)] Loss: 0.2711
INFO:__main__:Train Epoch: 4 [1984/6680 (30%)] Loss: 0.2760
INFO:__main__:Train Epoch: 4 [2016/6680 (30%)] Loss: 0.2793
INFO:__main__:Train Epoch: 4 [2048/6680 (31%)] Loss: 0.2825
INFO:__main__:Train Epoch: 4 [2080/6680 (31%)] Loss: 0.2873
INFO:__main__:Train Epoch: 4 [2112/6680 (32%)] Loss: 0.2904
INFO:__main__:Train Epoch: 4 [2144/6680 (32%)] Loss: 0.2935
INFO:__main__:Train Epoch: 4 [2176/6680 (33%)] Loss: 0.2974
INFO:__main__:Train Epoch: 4 [2208/6680 (33%)] Loss: 0.3026
INFO:__main__:Train Epoch: 4 [2240/6680 (34%)] Loss: 0.3067
INFO:__main__:Train Epoch: 4 [2272/6680 (34%)] Loss: 0.3123
INFO:__main__:Train Epoch: 4 [2304/6680 (34%)] Loss: 0.3169
INFO:__main__:Train Epoch: 4 [2336/6680 (35%)] Loss: 0.3218
INFO:__main__:Train Epoch: 4 [2368/6680 (35%)] Loss: 0.3265
INFO:__main__:Train Epoch: 4 [2400/6680 (36%)] Loss: 0.3305
```

```
INFO:__main__:Train Epoch: 4 [2432/6680 (36%)] Loss: 0.3346
INFO:__main__:Train Epoch: 4 [2464/6680 (37%)] Loss: 0.3400
INFO:__main__:Train Epoch: 4 [2496/6680 (37%)] Loss: 0.3437
INFO:__main__:Train Epoch: 4 [2528/6680 (38%)] Loss: 0.3480
INFO:__main__:Train Epoch: 4 [2560/6680 (38%)] Loss: 0.3520
INFO:__main__:Train Epoch: 4 [2592/6680 (39%)] Loss: 0.3558
INFO:__main__:Train Epoch: 4 [2624/6680 (39%)] Loss: 0.3596
INFO:__main__:Train Epoch: 4 [2656/6680 (40%)] Loss: 0.3629
INFO:__main__:Train Epoch: 4 [2688/6680 (40%)] Loss: 0.3672
INFO:__main__:Train Epoch: 4 [2720/6680 (41%)] Loss: 0.3706
INFO:__main__:Train Epoch: 4 [2752/6680 (41%)] Loss: 0.3749
INFO:__main__:Train Epoch: 4 [2784/6680 (42%)] Loss: 0.3787
INFO:__main__:Train Epoch: 4 [2816/6680 (42%)] Loss: 0.3831
INFO:__main__:Train Epoch: 4 [2848/6680 (43%)] Loss: 0.3888
INFO:__main__:Train Epoch: 4 [2880/6680 (43%)] Loss: 0.3915
INFO:__main__:Train Epoch: 4 [2912/6680 (44%)] Loss: 0.3949
INFO:__main__:Train Epoch: 4 [2944/6680 (44%)] Loss: 0.3981
INFO:__main__:Train Epoch: 4 [2976/6680 (45%)] Loss: 0.4027
INFO:__main__:Train Epoch: 4 [3008/6680 (45%)] Loss: 0.4074
INFO:__main__:Train Epoch: 4 [3040/6680 (46%)] Loss: 0.4114
INFO:__main__:Train Epoch: 4 [3072/6680 (46%)] Loss: 0.4158
INFO:__main__:Train Epoch: 4 [3104/6680 (46%)] Loss: 0.4204
INFO:__main__:Train Epoch: 4 [3136/6680 (47%)] Loss: 0.4229
INFO:__main__:Train Epoch: 4 [3168/6680 (47%)] Loss: 0.4268
INFO:__main__:Train Epoch: 4 [3200/6680 (48%)] Loss: 0.4310
INFO:__main__:Train Epoch: 4 [3232/6680 (48%)] Loss: 0.4348
INFO:__main__:Train Epoch: 4 [3264/6680 (49%)] Loss: 0.4397
INFO:__main__:Train Epoch: 4 [3296/6680 (49%)] Loss: 0.4442
INFO:__main__:Train Epoch: 4 [3328/6680 (50%)] Loss: 0.4483
INFO:__main__:Train Epoch: 4 [3360/6680 (50%)] Loss: 0.4519
INFO:__main__:Train Epoch: 4 [3392/6680 (51%)] Loss: 0.4558
INFO:__main__:Train Epoch: 4 [3424/6680 (51%)] Loss: 0.4595
INFO:__main__:Train Epoch: 4 [3456/6680 (52%)] Loss: 0.4632
INFO:__main__:Train Epoch: 4 [3488/6680 (52%)] Loss: 0.4675
INFO:__main__:Train Epoch: 4 [3520/6680 (53%)] Loss: 0.4711
INFO:__main__:Train Epoch: 4 [3552/6680 (53%)] Loss: 0.4756
INFO:__main__:Train Epoch: 4 [3584/6680 (54%)] Loss: 0.4795
INFO:__main__:Train Epoch: 4 [3616/6680 (54%)] Loss: 0.4856
INFO:__main__:Train Epoch: 4 [3648/6680 (55%)] Loss: 0.4891
INFO:__main__:Train Epoch: 4 [3680/6680 (55%)] Loss: 0.4962
INFO:__main__:Train Epoch: 4 [3712/6680 (56%)] Loss: 0.5008
INFO:__main__:Train Epoch: 4 [3744/6680 (56%)] Loss: 0.5067
INFO:__main__:Train Epoch: 4 [3776/6680 (57%)] Loss: 0.5105
INFO:__main__:Train Epoch: 4 [3808/6680 (57%)] Loss: 0.5144
INFO:__main__:Train Epoch: 4 [3840/6680 (57%)] Loss: 0.5193
INFO:__main__:Train Epoch: 4 [3872/6680 (58%)] Loss: 0.5232
INFO:__main__:Train Epoch: 4 [3904/6680 (58%)] Loss: 0.5275
INFO:__main__:Train Epoch: 4 [3936/6680 (59%)] Loss: 0.5315
INFO:__main__:Train Epoch: 4 [3968/6680 (59%)] Loss: 0.5355
INFO:__main__:Train Epoch: 4 [4000/6680 (60%)] Loss: 0.5403
INFO:__main__:Train Epoch: 4 [4032/6680 (60%)] Loss: 0.5438
INFO:__main__:Train Epoch: 4 [4064/6680 (61%)] Loss: 0.5484
INFO:__main__:Train Epoch: 4 [4096/6680 (61%)] Loss: 0.5528
INFO:__main__:Train Epoch: 4 [4128/6680 (62%)] Loss: 0.5577
INFO:__main__:Train Epoch: 4 [4160/6680 (62%)] Loss: 0.5625
INFO:__main__:Train Epoch: 4 [4192/6680 (63%)] Loss: 0.5664
INFO:__main__:Train Epoch: 4 [4224/6680 (63%)] Loss: 0.5708
INFO:__main__:Train Epoch: 4 [4256/6680 (64%)] Loss: 0.5741
INFO:__main__:Train Epoch: 4 [4288/6680 (64%)] Loss: 0.5789
INFO:__main__:Train Epoch: 4 [4320/6680 (65%)] Loss: 0.5826
INFO:__main__:Train Epoch: 4 [4352/6680 (65%)] Loss: 0.5857
INFO:__main__:Train Epoch: 4 [4384/6680 (66%)] Loss: 0.5901
INFO:__main__:Train Epoch: 4 [4416/6680 (66%)] Loss: 0.5949
INFO:__main__:Train Epoch: 4 [4448/6680 (67%)] Loss: 0.6011
INFO:__main__:Train Epoch: 4 [4480/6680 (67%)] Loss: 0.6037
INFO:__main__:Train Epoch: 4 [4512/6680 (68%)] Loss: 0.6074
INFO:__main__:Train Epoch: 4 [4544/6680 (68%)] Loss: 0.6105
INFO:__main__:Train Epoch: 4 [4576/6680 (69%)] Loss: 0.6144
INFO:__main__:Train Epoch: 4 [4608/6680 (69%)] Loss: 0.6184
INFO:__main__:Train Epoch: 4 [4640/6680 (69%)] Loss: 0.6227
INFO:__main__:Train Epoch: 4 [4672/6680 (70%)] Loss: 0.6274
INFO:__main__:Train Epoch: 4 [4704/6680 (70%)] Loss: 0.6313
INFO:__main__:Train Epoch: 4 [4736/6680 (71%)] Loss: 0.6358
INFO:__main__:Train Epoch: 4 [4768/6680 (71%)] Loss: 0.6408
INFO:__main__:Train Epoch: 4 [4800/6680 (72%)] Loss: 0.6447
INFO:__main__:Train Epoch: 4 [4832/6680 (72%)] Loss: 0.6476
INFO:__main__:Train Epoch: 4 [4864/6680 (73%)] Loss: 0.6517
INFO:__main__:Train Epoch: 4 [4896/6680 (73%)] Loss: 0.6561
INFO:__main__:Train Epoch: 4 [4928/6680 (74%)] Loss: 0.6606
INFO:__main__:Train Epoch: 4 [4960/6680 (74%)] Loss: 0.6669
INFO:__main__:Train Epoch: 4 [4992/6680 (75%)] Loss: 0.6706
INFO:__main__:Train Epoch: 4 [5024/6680 (75%)] Loss: 0.6745
INFO:__main__:Train Epoch: 4 [5056/6680 (76%)] Loss: 0.6794
INFO:__main__:Train Epoch: 4 [5088/6680 (76%)] Loss: 0.6855
```

```
INFO:__main__:Train Epoch: 4 [5120/6680 (77%)] Loss: 0.6892
INFO:__main__:Train Epoch: 4 [5152/6680 (77%)] Loss: 0.6927
INFO:__main__:Train Epoch: 4 [5184/6680 (78%)] Loss: 0.6964
INFO:__main__:Train Epoch: 4 [5216/6680 (78%)] Loss: 0.7008
INFO:__main__:Train Epoch: 4 [5248/6680 (79%)] Loss: 0.7041
INFO:__main__:Train Epoch: 4 [5280/6680 (79%)] Loss: 0.7067
INFO:__main__:Train Epoch: 4 [5312/6680 (80%)] Loss: 0.7113
INFO:__main__:Train Epoch: 4 [5344/6680 (80%)] Loss: 0.7163
INFO:__main__:Train Epoch: 4 [5376/6680 (80%)] Loss: 0.7214
INFO:__main__:Train Epoch: 4 [5408/6680 (81%)] Loss: 0.7250
INFO:__main__:Train Epoch: 4 [5440/6680 (81%)] Loss: 0.7295
INFO:__main__:Train Epoch: 4 [5472/6680 (82%)] Loss: 0.7328
INFO:__main__:Train Epoch: 4 [5504/6680 (82%)] Loss: 0.7365
INFO:__main__:Train Epoch: 4 [5536/6680 (83%)] Loss: 0.7406
INFO:__main__:Train Epoch: 4 [5568/6680 (83%)] Loss: 0.7450
INFO:__main__:Train Epoch: 4 [5600/6680 (84%)] Loss: 0.7511
INFO:__main__:Train Epoch: 4 [5632/6680 (84%)] Loss: 0.7561
INFO:__main__:Train Epoch: 4 [5664/6680 (85%)] Loss: 0.7622
INFO:__main__:Train Epoch: 4 [5696/6680 (85%)] Loss: 0.7657
INFO:__main__:Train Epoch: 4 [5728/6680 (86%)] Loss: 0.7692
INFO:__main__:Train Epoch: 4 [5760/6680 (86%)] Loss: 0.7735
INFO:__main__:Train Epoch: 4 [5792/6680 (87%)] Loss: 0.7781
INFO:__main__:Train Epoch: 4 [5824/6680 (87%)] Loss: 0.7804
INFO:__main__:Train Epoch: 4 [5856/6680 (88%)] Loss: 0.7833
INFO:__main__:Train Epoch: 4 [5888/6680 (88%)] Loss: 0.7865
INFO:__main__:Train Epoch: 4 [5920/6680 (89%)] Loss: 0.7896
INFO:__main__:Train Epoch: 4 [5952/6680 (89%)] Loss: 0.7953
INFO:__main__:Train Epoch: 4 [5984/6680 (90%)] Loss: 0.7984
INFO:__main__:Train Epoch: 4 [6016/6680 (90%)] Loss: 0.8016
INFO:__main__:Train Epoch: 4 [6048/6680 (91%)] Loss: 0.8057
INFO:__main__:Train Epoch: 4 [6080/6680 (91%)] Loss: 0.8092
INFO:__main__:Train Epoch: 4 [6112/6680 (91%)] Loss: 0.8141
INFO:__main__:Train Epoch: 4 [6144/6680 (92%)] Loss: 0.8175
INFO:__main__:Train Epoch: 4 [6176/6680 (92%)] Loss: 0.8212
INFO:__main__:Train Epoch: 4 [6208/6680 (93%)] Loss: 0.8264
INFO:__main__:Train Epoch: 4 [6240/6680 (93%)] Loss: 0.8301
INFO:__main__:Train Epoch: 4 [6272/6680 (94%)] Loss: 0.8326
INFO:__main__:Train Epoch: 4 [6304/6680 (94%)] Loss: 0.8357
INFO:__main__:Train Epoch: 4 [6336/6680 (95%)] Loss: 0.8408
INFO:__main__:Train Epoch: 4 [6368/6680 (95%)] Loss: 0.8437
INFO:__main__:Train Epoch: 4 [6400/6680 (96%)] Loss: 0.8469
INFO:__main__:Train Epoch: 4 [6432/6680 (96%)] Loss: 0.8502
INFO:__main__:Train Epoch: 4 [6464/6680 (97%)] Loss: 0.8536
INFO:__main__:Train Epoch: 4 [6496/6680 (97%)] Loss: 0.8585
INFO:__main__:Train Epoch: 4 [6528/6680 (98%)] Loss: 0.8638
INFO:__main__:Train Epoch: 4 [6560/6680 (98%)] Loss: 0.8680
INFO:__main__:Train Epoch: 4 [6592/6680 (99%)] Loss: 0.8714
INFO:__main__:Train Epoch: 4 [6624/6680 (99%)] Loss: 0.8744
INFO:__main__:Train Epoch: 4 [6656/6680 (100%)] Loss: 0.8786
INFO:__main__:Train Epoch: 4 [5016/6680 (75%)] Loss: 0.8839
INFO:__main__:Validate Epoch: 4 [32/835 (4%)] Loss: 0.0148
INFO:__main__:Validate Epoch: 4 [64/835 (8%)] Loss: 0.0421
INFO:__main__:Validate Epoch: 4 [96/835 (11%)] Loss: 0.0722
INFO:__main__:Validate Epoch: 4 [128/835 (15%)] Loss: 0.0866
INFO:__main__:Validate Epoch: 4 [160/835 (19%)] Loss: 0.0980
INFO:__main__:Validate Epoch: 4 [192/835 (23%)] Loss: 0.1247
INFO:__main__:Validate Epoch: 4 [224/835 (27%)] Loss: 0.1632
INFO:__main__:Validate Epoch: 4 [256/835 (31%)] Loss: 0.1942
INFO:__main__:Validate Epoch: 4 [288/835 (34%)] Loss: 0.2287
INFO:__main__:Validate Epoch: 4 [320/835 (38%)] Loss: 0.2686
INFO:__main__:Validate Epoch: 4 [352/835 (42%)] Loss: 0.3064
INFO:__main__:Validate Epoch: 4 [384/835 (46%)] Loss: 0.3339
INFO:__main__:Validate Epoch: 4 [416/835 (50%)] Loss: 0.3701
INFO:__main__:Validate Epoch: 4 [448/835 (54%)] Loss: 0.3936
INFO:__main__:Validate Epoch: 4 [480/835 (57%)] Loss: 0.4311
INFO:__main__:Validate Epoch: 4 [512/835 (61%)] Loss: 0.4709
INFO:__main__:Validate Epoch: 4 [544/835 (65%)] Loss: 0.5182
INFO:__main__:Validate Epoch: 4 [576/835 (69%)] Loss: 0.5622
INFO:__main__:Validate Epoch: 4 [608/835 (73%)] Loss: 0.5824
INFO:__main__:Validate Epoch: 4 [640/835 (77%)] Loss: 0.6062
INFO:__main__:Validate Epoch: 4 [672/835 (80%)] Loss: 0.6502
INFO:__main__:Validate Epoch: 4 [704/835 (84%)] Loss: 0.6910
INFO:__main__:Validate Epoch: 4 [736/835 (88%)] Loss: 0.7144
INFO:__main__:Validate Epoch: 4 [768/835 (92%)] Loss: 0.7464
INFO:__main__:Validate Epoch: 4 [800/835 (96%)] Loss: 0.7907
INFO:__main__:Validate Epoch: 4 [832/835 (100%)] Loss: 0.8274
INFO:__main__:Validate Epoch: 4 [81/835 (10%)] Loss: 0.8826
INFO:__main__:Saving the model.
```

2024-08-17 09:08:49 Completed - Training job completed

Training seconds: 1324

Billable seconds: 1324


```
In [40... # TODO: Plot a debugging output.
job_name = estimator.latest_training_job.name
client = estimator.sagemaker_session.sagemaker_client
description = client.describe_training_job(TrainingJobName=estimator.latest_training_job.name)
```

```
In [41... from smdebug.trials import create_trial
from smdebug.core.modes import ModeKeys
```

```
trial = create_trial(estimator.latest_job_debugger_artifacts_path())
trial.tensor_names()
```

[2024-08-17 09:09:37.785 ip-172-16-46-62.ec2.internal:12032 INFO s3_trial.py:42] Loading trial debug-output at path s3://sagemaker-us-eas-t-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/debug-output

[2024-08-17 09:09:39.551 ip-172-16-46-62.ec2.internal:12032 INFO trial.py:197] Training has ended, will refresh one final time in 1 sec.

[2024-08-17 09:09:40.568 ip-172-16-46-62.ec2.internal:12032 INFO trial.py:210] Loaded all steps

```
Out[41]: ['CrossEntropyLoss_output_0',
'gradient/ResNet_fc.0.bias',
'gradient/ResNet_fc.0.weight',
'gradient/ResNet_fc.2.bias',
'gradient/ResNet_fc.2.weight',
'layer1.0.relu_input_0',
'layer1.0.relu_input_1',
'layer1.1.relu_input_0',
'layer1.1.relu_input_1',
'layer2.0.relu_input_0',
'layer2.0.relu_input_1',
'layer2.1.relu_input_0',
'layer2.1.relu_input_1',
'layer3.0.relu_input_0',
'layer3.0.relu_input_1',
'layer3.1.relu_input_0',
'layer3.1.relu_input_1',
'layer4.0.relu_input_0',
'layer4.0.relu_input_1',
'layer4.1.relu_input_0',
'layer4.1.relu_input_1',
'relu_input_0']
```

```
In [42... print(len(trial.tensor("CrossEntropyLoss_output_0").steps(mode=ModeKeys.TRAIN)))
print(len(trial.tensor("CrossEntropyLoss_output_0").steps(mode=ModeKeys.EVAL)))
```

```
9
22
```

```
In [43... def get_data(trial, tname, mode):
    tensor = trial.tensor(tname)
    steps = tensor.steps(mode=mode)
    vals = []
    for s in steps:
        vals.append(tensor.value(s, mode=mode))
    return steps, vals
```

```
In [44... import matplotlib.pyplot as plt
from mpl_toolkits.axes_grid1 import host_subplot

def plot_tensor(trial, tensor_name):

    steps_train, vals_train = get_data(trial, tensor_name, mode=ModeKeys.TRAIN)
    print("loaded TRAIN data")
    steps_eval, vals_eval = get_data(trial, tensor_name, mode=ModeKeys.EVAL)
    print("loaded EVAL data")

    fig = plt.figure(figsize=(10, 7))
    host = host_subplot(111)

    par = host.twinx()

    host.set_xlabel("Steps (TRAIN)")
    par.set_xlabel("Steps (EVAL)")
    host.set_ylabel(tensor_name)

    (p1,) = host.plot(steps_train, vals_train, label=tensor_name)
    print("completed TRAIN plot")
    (p2,) = par.plot(steps_eval, vals_eval, label="val_" + tensor_name)
    print("completed EVAL plot")
    leg = plt.legend()

    host.xaxis.get_label().set_color(p1.get_color())
    leg.texts[0].set_color(p1.get_color())

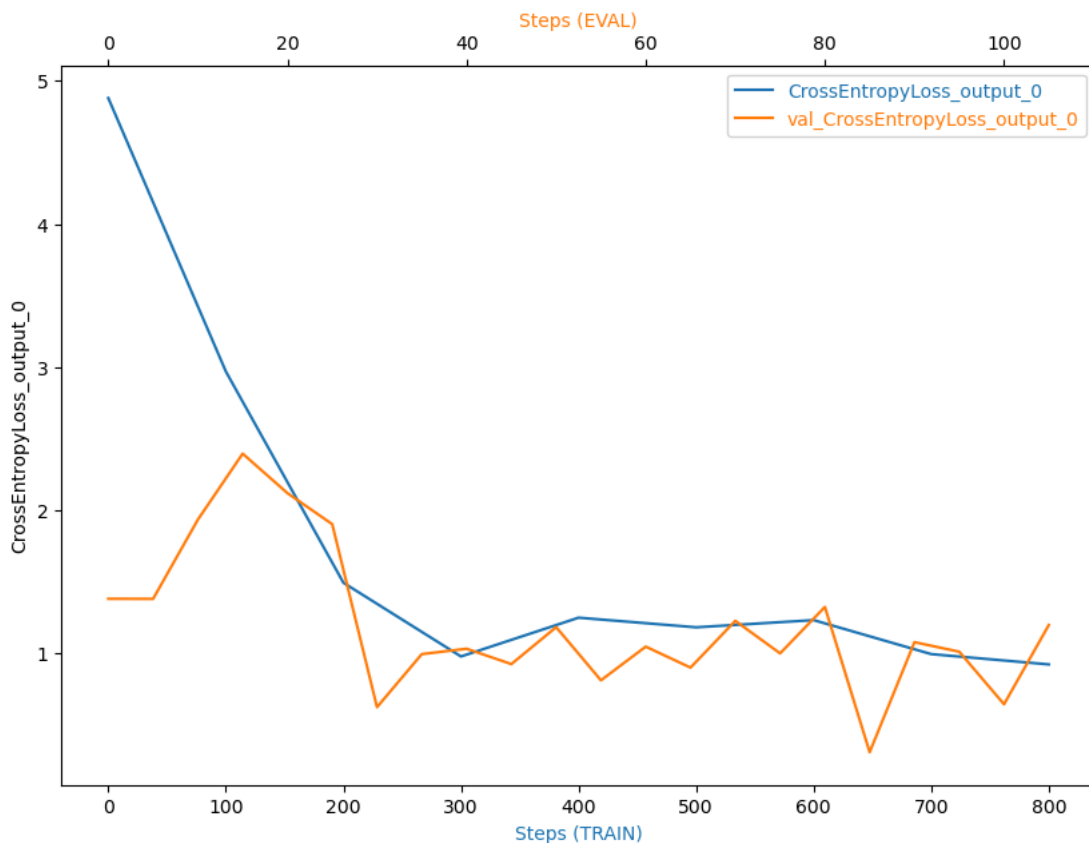
    par.xaxis.get_label().set_color(p2.get_color())
    leg.texts[1].set_color(p2.get_color())

    plt.ylabel(tensor_name)

    plt.show()
```

```
In [45... plot_tensor(trial, "CrossEntropyLoss_output_0")
```

```
loaded TRAIN data
loaded EVAL data
completed TRAIN plot
completed EVAL plot
```



TODD: Is there some anomalous behaviour in your debugging output? If so, what is the error and how will you fix it?

TODD: If not, suppose there was an error. What would that error look like and how would you have fixed it?


```
In [46... # TODO: Display the profiler output
rule_output_path = estimator.output_path + estimator.latest_training_job.job_name + "/rule-output"
print(f"You will find the profiler report in {rule_output_path}")
```

You will find the profiler report in s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output

```
In [47... ! aws s3 ls {rule_output_path} --recursive
```

```
2024-08-17 09:08:38      375469 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-report.html
2024-08-17 09:08:37      223399 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-report.ipynb
2024-08-17 09:08:32         192 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/Batch
Size.json
2024-08-17 09:08:32         200 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/CPUBo
ttleneck.json
2024-08-17 09:08:32       1970 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/DataL
oader.json
2024-08-17 09:08:32        127 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/GPUMe
moryIncrease.json
2024-08-17 09:08:32        199 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/IOBot
tleneck.json
2024-08-17 09:08:32        119 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/LoadB
alancing.json
2024-08-17 09:08:32        151 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/LowGP
UUtilization.json
2024-08-17 09:08:32        232 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/MaxIn
itializationTime.json
2024-08-17 09:08:32       1271 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/Overa
llFrameworkMetrics.json
2024-08-17 09:08:32         473 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/Overa
llSystemUsage.json
2024-08-17 09:08:32       2277 pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profiler-reports/StepO
utlier.json
```

```
In [48... ! aws s3 cp {rule_output_path} ./ --recursive
```

```
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/BatchSize.json to ProfilerReport/profiler-output/profiler-reports/BatchSize.json
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/Dataloader.json to ProfilerReport/profiler-output/profiler-reports/Dataloader.json
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/OverallSystemUsage.json to ProfilerReport/profiler-output/profiler-reports/OverallSystemUsage.json
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/IOBottleneck.json to ProfilerReport/profiler-output/profiler-reports/IOBottleneck.json
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/MaxInitializationTime.json to ProfilerReport/profiler-output/profiler-reports/MaxInitializationTime.json
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/OverallFrameworkMetrics.json to ProfilerReport/profiler-output/profiler-reports/OverallFrameworkMetrics.json
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-report.html to ProfilerReport/profiler-output/profiler-report.html
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/LowGPUUtilization.json to ProfilerReport/profiler-output/profiler-reports/LowGPUUtilization.json
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/LoadBalancing.json to ProfilerReport/profiler-output/profiler-reports/LoadBalancing.json
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/StepOutlier.json to ProfilerReport/profiler-output/profiler-reports/StepOutlier.json
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-report.ipynb to ProfilerReport/profiler-output/profiler-report.ipynb
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/CPUBottleneck.json to ProfilerReport/profiler-output/profiler-reports/CPUBottleneck.json
download: s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/rule-output/ProfilerReport/profiler-output/profi
ler-reports/GPUMemoryIncrease.json to ProfilerReport/profiler-output/profiler-reports/GPUMemoryIncrease.json
```

```
In [19... import os

# get the autogenerated folder name of profiler report
profiler_report_name = [
    rule["RuleConfigurationName"]
    for rule in estimator.latest_training_job.rule_job_summary()
    if "Profiler" in rule["RuleConfigurationName"]][0]
```

```
In [49... import IPython

IPython.display.HTML(filename="./ProfilerReport/profiler-output/profiler-report.html")
```

Out[49]:

SageMaker Debugger Profiling Report

SageMaker Debugger auto generated this report. You can generate similar reports on all supported training jobs. The report provides summary of training job metrics, rules summary, and detailed analysis from each rule. The graphs and tables are interactive.

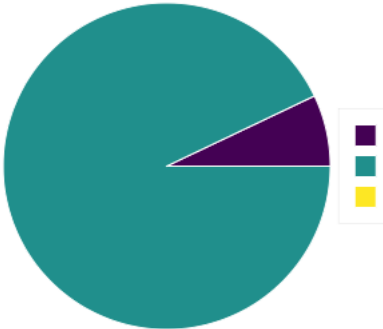
Legal disclaimer: This report and any recommendations are provided for informational purposes only and are not definitive. You are responsible for making your own information.

```
In [4]: # Parameters
processing_job_arn = "arn:aws:sagemaker:us-east-1:065326103865:processing-job/pytorch-training-2024-08-1-ProfilerReport-e871398b"
```

Training job summary

The following table gives a summary about the training job. The table includes information about when the training job started and ended, how much time initialization, training loop and finalization took. Your training job started on 08/17/2024 at 08:46:32 and ran for 1287 seconds.

#		Job Statistics
0	Start time	08:46:32 08/17/2024
1	End time	09:07:59 08/17/2024
2	Job duration	1287 seconds
3	Training loop start	08:48:05 08/17/2024
4	Training loop end	09:07:58 08/17/2024
5	Training loop duration	1193 seconds
6	Initialization time	93 seconds
7	Finalization time	0 seconds
8	Initialization	7 %
9	Training loop	92 %
10	Finalization	0 %



System usage statistics

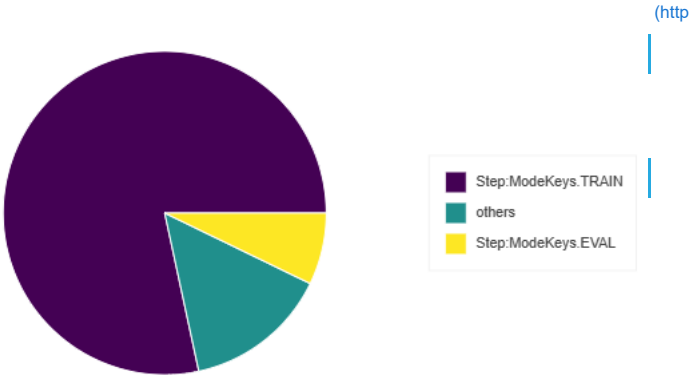
The following table shows statistics of resource utilization per worker (node), such as the total CPU and GPU utilization, and the memory utilization on CPU and GPU. The table also includes the total I/O wait time and the total amount of data sent or received in bytes. The table shows min and max values as well as p99, p90 and p50 percentiles.

#	node	metric	unit	max	p99	p95	p50	min
0	algo-1	Network	bytes	49937250.54	0	0	0	0
1	algo-1	CPU	percentage	99.46	94.52	70.6	49.02	0.98
2	algo-1	CPU memory	percentage	15.5	14.12	13.84	13.07	3.27
3	algo-1	I/O	percentage	27.16	12.18	1.55	0	0

Framework metrics summary

The following two pie charts show the time spent on the TRAIN phase, the EVAL phase, and others. The 'others' includes the time spent between steps the next step has started). Ideally, most of the training time should be spent on the TRAIN and EVAL phases. If TRAIN/EVAL were not specified in the tr GLOBAL.

The ratio between the time spent on the TRAIN/EVAL phase and others

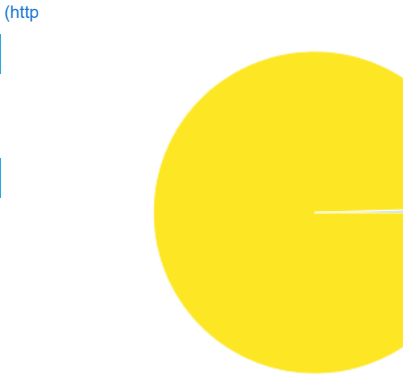


The following piechart shows a breakdown of the CPU/GPU operators. It shows that 100% of training time was spent on executing the "cpu_functions" c

The ratio between the time spent on CPU/GPU operators



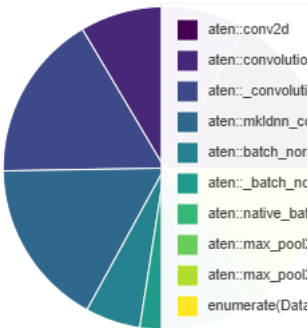
General framework operations



Overview: CPU operators

The following table shows a list of operators that ran on the CPUs. The most expensive operator on the CPUs was "aten::conv2d" with 16 %.

#	Percentage	Cumulative time in microseconds	CPU operator
0	16.76	6319677	aten::conv2d
1	16.76	6318463	aten::convolution
2	16.75	6317398	aten::_convolution
3	16.75	6314846	aten::mkldnn_convolution
4	11.12	4193724	enumerate(DataLoader)#_Singlef
5	5.51	2078800	aten::batch_norm
6	5.51	2077385	aten::_batch_norm_impl_index
7	5.5	2075517	aten::native_batch_norm
8	2.67	1008665	aten::max_pool2d
9	2.66	1002783	aten::max_pool2d_with_indices



Rules summary

The following table shows a profiling summary of the Debugger built-in rules. The table is sorted by the rules that triggered the most frequent rule was the most frequently triggered. It processed 0 datapoints and was triggered 0 times.

		Description	Recommendation	N
LowGPUUtilization	Checks if the GPU utilization is low or fluctuating. This can happen due to bottlenecks, blocking calls for synchronizations, or a small batch size.	Check if there are bottlenecks, minimize blocking calls, change distributed training strategy, or increase the batch size.		
Dataloader	Checks how many data loaders are running in parallel and whether the total number is equal the number of available CPU cores. The rule triggers if number is much smaller or larger than the number of available cores. If too small, it might lead to low GPU utilization. If too large, it might impact other compute intensive operations on CPU.	Change the number of data loader processes.		
GPUMemoryIncrease	Measures the average GPU memory footprint and triggers if there is a large increase.	Choose a larger instance type with more memory if footprint is close to maximum available memory.		
MaxInitializationTime	Checks if the time spent on initialization exceeds a threshold percent of the total training time. The rule waits until the first step of training loop starts. The initialization can take longer if downloading the entire dataset from Amazon S3 in File mode. The default threshold is 20 minutes.	Initialization takes too long. If using File mode, consider switching to Pipe mode in case you are using TensorFlow framework.		
LoadBalancing	Detects workload balancing issues across GPUs. Workload imbalance can occur in training jobs with data parallelism. The gradients are accumulated on a primary GPU, and this GPU might be overused with regard to other GPUs, resulting in reducing the efficiency of data parallelization.	Choose a different distributed training strategy or a different distributed training framework.		
BatchSize	Checks if GPUs are underutilized because the batch size is too small. To detect this problem, the rule analyzes the average GPU memory footprint, the CPU and the GPU utilization.	The batch size is too small, and GPUs are underutilized. Consider running on a smaller instance type or increasing the batch size.		
CPUBottleneck	Checks if the CPU utilization is high and the GPU utilization is low. It might indicate CPU bottlenecks, where the GPUs are waiting for data to arrive from the CPUs. The rule evaluates the CPU and GPU utilization rates, and triggers the issue if the time spent on the CPU bottlenecks exceeds a threshold percent of the total training time. The default threshold is 50 percent.	Consider increasing the number of data loaders or applying data pre-fetching.		
IOBottleneck	Checks if the data I/O wait time is high and the GPU utilization is low. It might indicate IO bottlenecks where GPU is waiting for data to arrive from storage. The rule evaluates the I/O and GPU utilization rates and triggers the issue if the time spent on the IO bottlenecks exceeds a threshold percent of the total training time. The default threshold is 50 percent.	Pre-fetch data or choose different file formats, such as binary formats that improve I/O performance.		
StepOutlier	Detects outliers in step duration. The step duration for forward and backward pass should be roughly the same throughout the training. If there are significant outliers, it may indicate a system stall or bottleneck issues.	Check if there are any bottlenecks (CPU, I/O) correlated to the step outliers.		

Analyzing the training loop

Step duration analysis

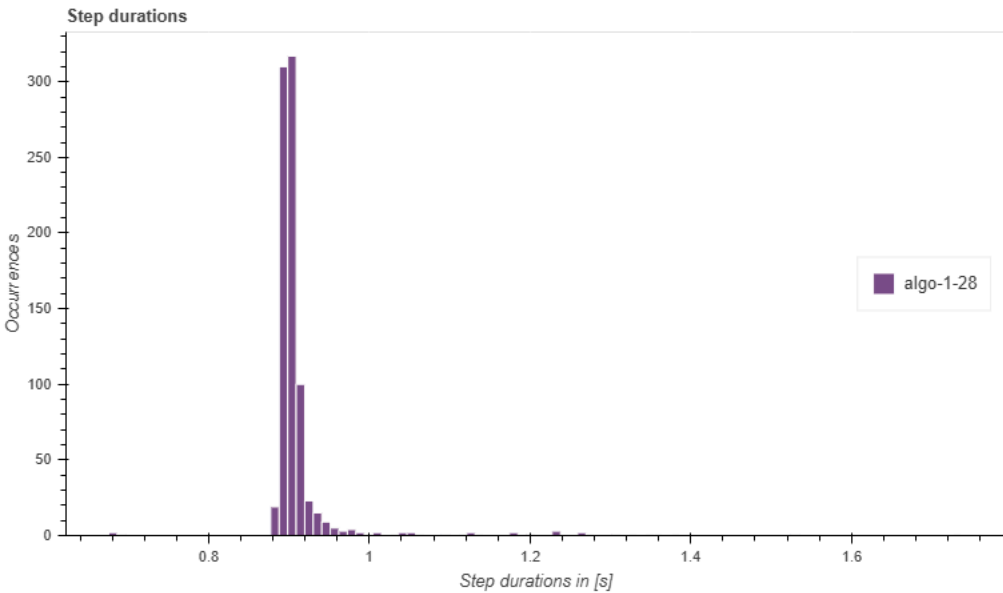
The StepOutlier rule measures step durations and checks for outliers. The rule returns True if duration is larger than 3 times the standard deviation. The also takes the parameter mode, that specifies whether steps from training or validation phase should be checked. In your processing job mode was spe as None. Typically the first step is taking significantly more time and to avoid the rule triggering immediately, one can use n_outliers to specify the numb outliers to ignore. n_outliers was set to 10. The rule analysed 942 datapoints and triggered 0 times.

Step durations on node algo-1-28:

The following table is a summary of the statistics of step durations measured on node algo-1-28. The rule has analyzed the step duration from duration on node algo-1-28 was 0.96s. The rule detected 10 outliers, where step duration was larger than 3 times the standard deviation of 0.4

	mean	max	p99	p95	p50	min
Step Durations in [s]	0.96	5.48	3.73	0.98	0.90	0.68

The following histogram shows the step durations measured on the different nodes. You can turn on or turn off the visualization of histograms:



(http
|

GPU utilization analysis

Usage per GPU

The LowGPUUtilization rule checks for a low and fluctuating GPU usage. If the GPU usage is consistently low, it might be caused by bottlenecks or a small batch size. If usage is heavily fluctuating, it can be due to bottlenecks or blocking calls. The rule computed the 95th and 5th percentile of GPU utilization on 500 continuous datapoints and found 0 cases where p95 was above 70% and p5 was below 10%. If p95 is high and p5 is low, it might indicate that the GPU usage is highly fluctuating. If both values are very low, it would mean that the machine is underutilized. During initialization, the GPU usage is likely zero, so the rule skipped the first 1000 data points. The rule analysed 0 datapoints and triggered 0 times.

Workload balancing

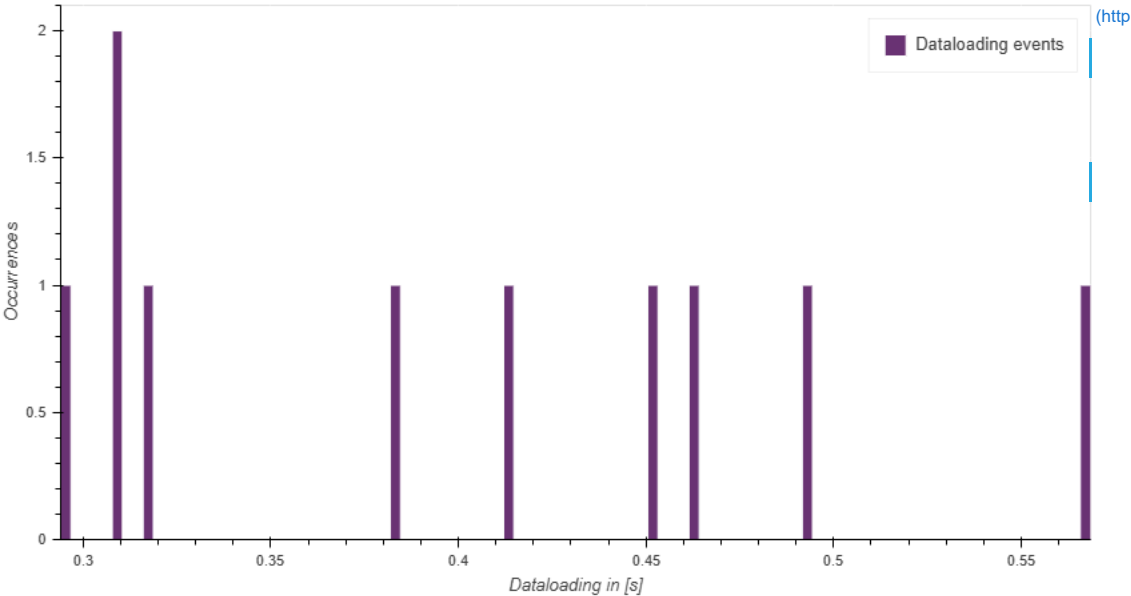
The LoadBalancing rule helps to detect issues in workload balancing between multiple GPUs. It computes a histogram of GPU utilization values for eac GPU and compares then the similarity between histograms. The rule checked if the distance of histograms is larger than the threshold of 0.2. During initialization utilization is likely zero, so the rule skipped the first 1000 data points.

Dataloading analysis

The number of dataloader workers can greatly affect the overall performance of your training job. The rule analyzed the number of dataloading processes that have been running in parallel on the training instance and compares it against the total number of cores. The rule checked if the number of process smaller than 70% or larger than 200% the total number of cores. Having too few dataloader workers can slowdown data preprocessing and lead to GPU underutilization. Having too many dataloader workers may hurt the overall performance if you are running other compute intensive tasks on the CPU. TI rule analysed 10 datapoints and triggered 0 times.

Your training instance provided 4 CPU cores, however your training job only ran on average 1 dataloader workers in parallel. We recommend you to increase the number of dataloader workers. Using pinned memory also improves performance because it enables fast data transfer to CUDA-enabled GPUs. The rule detected that your training job was not using pinned memory. In case of using PyTorch Dataloader, you can enable this by setting `pin_memory=True`.

The following histogram shows the distribution of dataloading times that have been measured throughout your training job. The median dataloading time was 0.3981s. The 95th percentile was 0.5347s and the 25th percentile was 0.312s



```
In [50... import boto3

session = boto3.session.Session()
region = session.region_name

training_job_name = estimator.latest_training_job.name
print(f"Training jobname: {training_job_name}")

Training jobname: pytorch-training-2024-08-17-08-45-48-485
```

```
In [51... from smdebug.profiler.analysis.notebook_utils.training_job import TrainingJob

tj = TrainingJob(training_job_name, region)
tj.wait_for_sys_profiling_data_to_be_available()

ProfilerConfig: {'S3OutputPath': 's3://sagemaker-us-east-1-065326103865/', 'ProfilingIntervalInMilliseconds': 500, 'ProfilingParameters':
{'DataloaderProfilingConfig': '{"StartStep": 0, "NumSteps": 10, "MetricsRegex": ".*", }', 'DetailedProfilingConfig': '{"StartStep": 0, "N
umSteps": 10, }', 'FileOpenFailThreshold': '50', 'HorovodProfilingConfig': '{"StartStep": 0, "NumSteps": 10, }', 'LocalPath': '/opt/ml/ou
tput/profiler', 'PythonProfilingConfig': '{"StartStep": 0, "NumSteps": 10, "ProfilerName": "cprofile", "cProfileTimer": "total_time", }',
'RotateFileCloseIntervalInSeconds': '60', 'RotateMaxFileSizeInBytes': '10485760', 'SMDDataParallelProfilingConfig': '{"StartStep": 0, "Num
Steps": 10, }', 'DisableProfiler': False}
s3 path:s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-08-45-48-485/profiler-output

Profiler data from system is available
```

```
In [52... from smdebug.profiler.analysis.notebook_utils.timeline_charts import TimelineCharts

system_metrics_reader = tj.get_systems_metrics_reader()
system_metrics_reader.refresh_event_file_list()

view_timeline_charts = TimelineCharts(
    system_metrics_reader,
    framework_metrics_reader=None,
    select_dimensions=["CPU", "GPU"],
    select_events=["total"],
)
```

```
[2024-08-17 09:11:40.548 ip-172-16-46-62.ec2.internal:12032 INFO metrics_reader_base.py:134] Getting 23 event files
select events:['total']
select dimensions:['CPU', 'GPU']
filtered_events: {'total'}
filtered_dimensions: {'CPUUtilization-nodeid:algo-1'}
```

Model Deploying

```
In [32... # TODO: Deploy your model to an endpoint

# TODO: Add your deployment configuration like instance type and number of instances

predictor = estimator.deploy(initial_instance_count=1, instance_type="ml.m5.xlarge")
```

```
INFO:sagemaker:Repacking model artifact (s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-01-09-33-139/output/model.tar.gz), script artifact (s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-01-09-33-139/source/sourcedir.tar.gz), and dependencies ([]) into single tar.gz file located at s3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-02-35-22-730/model.tar.gz. This may take some time depending on model size...
INFO:sagemaker:Creating model with name: pytorch-training-2024-08-17-02-35-22-730
INFO:sagemaker:Creating endpoint-config with name pytorch-training-2024-08-17-02-35-22-730
INFO:sagemaker:Creating endpoint with name pytorch-training-2024-08-17-02-35-22-730
-----!
```

```
In [1]: import json
import boto3
import sagemaker
from sagemaker import get_execution_role
from IPython.core.display import Image, display
from sagemaker.pytorch import PyTorchModel

sagemaker_session = sagemaker.Session()
boto3 = sagemaker_session.boto_session
s3 = boto3.client("s3")
role = sagemaker.get_execution_role()
bucket = sagemaker_session.default_bucket()

model_data = 's3://sagemaker-us-east-1-065326103865/pytorch-training-2024-08-17-02-35-22-730/model.tar.gz'
image = '763104351884.dkr.ecr.us-east-1.amazonaws.com/pytorch-inference:1.8-cpu-py36'
model_name = 'pytorch-training-2024-08-17-02-35-22-730'

sagemaker.config INFO - Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
sagemaker.config INFO - Not applying SDK defaults from location: /home/ec2-user/.config/sagemaker/config.yaml
/tmp/ipykernel_14328/2273389406.py:5: DeprecationWarning: Importing display from IPython.core.display is deprecated since IPython 7.14, please import from IPython display
  from IPython.core.display import Image, display
```

```
In [17... pytorch_model = PyTorchModel(model_data = model_data,
                               role=role,
                               entry_point='inference.py',
                               py_version='py36',
                               framework_version='1.8')

predictor=pytorch_model.deploy(initial_instance_count=1, instance_type='ml.m5.large')

-----!
```

```
In [24... import random

def generate_test_case():
    # Setup s3 in boto3
    s3 = boto3.resource('s3')

    # Randomly pick from sfn or test folders in our bucket
    objects = s3.Bucket(bucket).objects.filter(Prefix='project/data/test')

    # Grab any random object key from that folder!
    obj = random.choice([x.key for x in objects])

    return obj

pick = generate_test_case()

print(pick.split('/')[3])
print(pick.split('/')[4])

046.Cavalier_king_charles_spaniel
Cavalier_king_charles_spaniel_03291.jpg
```

```
In [35... # TODO: Run an prediction on the endpoint

from sagemaker.predictor import Predictor
from sagemaker.serializers import IdentitySerializer

predictor.serializer = IdentitySerializer('image/jpeg')

pick = generate_test_case()

s3.download_file(bucket, pick, pick.split('/')[4])

with open(pick.split('/')[4], "rb") as file:
    body = file.read()
    response = predictor.predict(body)
```

```
In [26... import numpy as np

file_list = sagemaker_session.list_s3_files(bucket=bucket, key_prefix='project/data/test')
class_list = np.unique([file.split('/')[3] for file in file_list])
```

```
In [27... print(class_list)
```



```
[ '001.Affenpinscher' '002.Afghan_hound' '003.Airedale_terrier' '004.Akita'
'005.Alaskan_malamute' '006.American_eskimo_dog' '007.American_foxhound'
'008.American_staffordshire_terrier' '009.American_water_spaniel'
'010.Anatolian_shepherd_dog' '011.Australian_cattle_dog'
'012.Australian_shepherd' '013.Australian_terrier' '014.Basenji'
'015.Basset_hound' '016.Beagle' '017.Bearded_collie' '018.Beauceron'
'019.Bedlington_terrier' '020.Belgian_malinois' '021.Belgian_sheepdog'
'022.Belgian_tervuren' '023.Bernese_mountain_dog' '024.Bichon_frise'
'025.Black_and_tan_coonhound' '026.Black_russian_terrier'
'027.Bloodhound' '028.Bluetick_coonhound' '029.Border_collie'
'030.Border_terrier' '031.Borzoi' '032.Boston_terrier'
'033.Bouvier_des_flandres' '034.Boxer' '035.Boykin_spaniel' '036.Briard'
'037.Brittany' '038.Brussels_griffon' '039.Bull_terrier' '040.Bulldog'
'041.Bullmastiff' '042.Cairn_terrier' '043.Canaan_dog' '044.Cane_corso'
'045.Cardigan_welsh_corgi' '046.Cavalier_king_charles_spaniel'
'047.Chesapeake_bay_retriever' '048.Chihuahua' '049.Chinese_crested'
'050.Chinese_shar_pei' '051.Chow_chow' '052.Clumber_spaniel'
'053.Cocker_spaniel' '054.Collie' '055.Curly-coated_retriever'
'056.Dachshund' '057.Dalmatian' '058.Dandie_dinmont_terrier'
'059.Doberman_pinscher' '060.Dogue_de_bordeaux'
'061.English_cocker_spaniel' '062.English_setter'
'063.English_springer_spaniel' '064.English_toy_spaniel'
'065.Entlebucher_mountain_dog' '066.Field_spaniel' '067.Finnish_spitz'
'068.Flat-coated_retriever' '069.French_bulldog' '070.German_pinscher'
'071.German_shepherd_dog' '072.German_shorthaired_pointer'
'073.German_wirehaired_pointer' '074.Giant_schnauzer'
'075.Glen_of_imaal_terrier' '076.Golden_retriever' '077.Gordon_setter'
'078.Great_dane' '079.Great_pyrenees' '080.Greater_swiss_mountain_dog'
'081.Greyhound' '082.Havanese' '083.Ibizan_hound'
'084.Icelandic_sheepdog' '085.Irish_red_and_white_setter'
'086.Irish_setter' '087.Irish_terrier' '088.Irish_water_spaniel'
'089.Irish_wolfhound' '090.Italian_greyhound' '091.Japanese_chin'
'092.Keeshond' '093.Kerry_blue_terrier' '094.Komondor' '095.Kuvasz'
'096.Labrador_retriever' '097.Lakeland_terrier' '098.Leonberger'
'099.Lhasa_apso' '100.Lowchen' '101.Maltese' '102.Manchester_terrier'
'103.Mastiff' '104.Minature_schnauzer' '105.Neapolitan_mastiff'
'106.Newfoundland' '107.Norfolk_terrier' '108.Norwegian_buhund'
'109.Norwegian_elkhound' '110.Norwegian_lundehund' '111.Norwich_terrier'
'112.Nova_scotia_duck_tolling_retriever' '113.Old_english_sheepdog'
'114.Otterhound' '115.Papillon' '116.Parson_russell_terrier'
'117.Pekingese' '118.Pembroke_welsh_corgi'
'119.Petit_basset_griffon_vendeen' '120.Pharao_hound' '121.Plott'
'122.Pointer' '123.Pomeranian' '124.Poodle' '125.Portuguese_water_dog'
'126.Saint_bernard' '127.Silky_terrier' '128.Smooth_fox_terrier'
'129.Tibetan_mastiff' '130.Welsh_springer_spaniel'
'131.Wirehaired_pointing_griffon' '132.Xoloitzcuintli'
'133.Yorkshire_terrier']
```

```
In [36... display(Image(filename=pick.split('/')[4]))
print('Image lable is {}'.format(pick.split('/')[3]))
print('Prediction is {}'.format(class_list[response.argmax()])))
```



Image lable is 023.Bernese_mountain_dog
Prediction is 023.Bernese_mountain_dog

```
In [ ]: # TODO: Remember to shutdown/delete your endpoint once your work is done
predictor.delete_endpoint()
```

```
In [ ]:
```